

07 Jul 2026 05:00:00 ET | 62 pages

Space Exploration Technologies (SPCX.O)

Potential for \$900+; Initiating at Buy

CITI'S TAKE

We are initiating on SPCX with a Buy rating and \$200 price target, implying ~25% upside by year-end. We believe this price target is a milestone along the path to \$900+ which becomes realistic assuming key engineering achievements are demonstrated at scale. Ultimately, the successful deployment of Starship will establish the most affordable and scalable path to unlocking the economic potential of space, in our view, giving SPCX access to trillion-dollar markets that no other company can realistically tap into. SPCX is at the beginning of a particularly catalyst-rich 2-3 years which we expect will result in the company establishing dominant positions in multiple high-value markets – namely Connectivity and AI. Join us for 10+ hours of webcasts on The Space Economy and Future of AI on July 8th.

Our bullish view is rooted in the following key points:

- Unrivaled **Launch** capabilities for the foreseeable future.
- Making the company uniquely capable of scaling space infrastructure and unlocking the trillion-dollar market opportunities it presents, including:
 - a) **Connectivity**: A Dominant Global Telecom Business.
 - b) **Enterprise & Orbital AI**: A Dominant Artificial Intelligence Business.
- Amplified by the benefits of **Extreme Vertical Integration**, which reinforces the company's ability to drive (a) costs down, (b) throughput up, and (c) deliver at a scale which other companies cannot replicate, in our view.

Valuation: Our \$200 YE26 price target is derived using an average of three valuation methodologies which we detail on page 4. Our \$900+ long-term valuation level is based on an illustrative framework that gives SPCX credit for supernormal growth rates continuing at scale; thus, sustaining a premium to *The Trillion Dollar Comps*.

Risks: Technology, manufacturing, and operating execution risks, as well as regulatory risks within each vertical. Further detail in each section of this report.

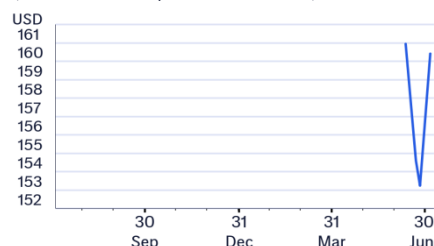
EPS (US\$)	Q1	Q2	Q3	Q4	FY	FC Cons
2025A	-0.14A	-0.26A	-0.27A	-0.60A	-1.27A	na
2026E	-1.10A	-0.40E	0.13E	0.25E	-0.24E	-0.96E
Previous	na	na	na	na	na	na
2027E	0.28E	0.20E	0.11E	0.15E	0.74E	0.11E
Previous	na	na	na	na	na	na
2028E	na	na	na	na	6.12E	0.70E
Previous	na	na	na	na	na	na

Source: Company Reports and dataCentral, Citi Research. FC Cons: First Call Consensus.

Buy

Price (06 Jul 26 16:00)	US\$160.42
Target price	US\$200.00
Expected share price return	24.7%
Expected dividend yield	0.0%
Expected total return	24.7%
Market Cap	US\$2,110,999M

Price Performance (RIC: SPCX.O, BB: SPCX US)



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SPCX.O: Fiscal year end 31-Dec						Price: US\$160.42; TP: US\$200.00; Market Cap: US\$2,110,999m; Recomm: Buy					
Profit & Loss (US\$m)	2024	2025	2026E	2027E	2028E	Valuation ratios	2024	2025	2026E	2027E	2028E
Sales revenue	na	18,674	38,655	69,926	163,318	PE (x)	na	na	na	na	26.2
Cost of sales	na	-9,451	-15,968	-26,280	-54,827	PB (x)	na	15.1	18.6	15.4	9.8
Gross profit	na	9,223	22,687	43,647	108,491	EV/EBITDA (x)	na	na	na	52.9	19.7
Gross Margin (%)	na	49.4	58.7	62.4	66.4	FCF yield (%)	na	-2.2	-2.3	-4.3	-4.2
EBITDA (Adj)	na	6,584	17,550	40,150	112,315	Dividend yield (%)	na	na	na	na	na
EBITDA Margin (Adj) (%)	na	35.3	45.4	57.4	68.8	Payout ratio (%)	na	0	0	0	0
Depreciation	na	-6,701	-13,181	-23,744	-37,127	ROE (%)	na	na	-2.6	7.8	45.6
Amortisation	na	0	0	0	0	Cashflow (US\$m)	2024	2025	2026E	2027E	2028E
EBIT (Adj)	na	-2,589	1,465	13,122	71,409	EBITDA	na	4,112	14,646	36,865	108,536
EBIT Margin (Adj) (%)	na	-13.9	3.8	18.8	43.7	Working capital	na	2,089	2,305	2,997	-20,007
Net interest	na	-1,453	-1,576	-3,257	-6,198	Other	na	584	456	-938	10,752
Associates	na	0	0	0	0	Operating cashflow	na	6,785	17,406	38,925	99,281
Non-Op/Except/Other Adj	na	-177	-1,876	0	0	Capex	na	-20,619	-48,046	-127,618	-185,922
Pre-tax profit	na	-4,219	-1,987	9,865	65,211	Net acq/disposals	na	0	0	0	0
Tax	na	-718	-6	-262	14,202	Other	na	1,044	-7,812	0	0
Extraord./Min.Int./Pref.div.	na	0	0	0	0	Investing cashflow	na	-19,575	-55,858	-127,618	-185,922
Reported net profit	na	-4,937	-1,993	9,603	79,413	Dividends paid	na	0	0	0	0
Net Margin (%)	na	-26.4	-5.2	13.7	48.6	Financing cashflow	na	26,350	71,552	76,045	119,684
Core NPAT	na	-4,937	-1,993	9,603	79,413	Net change in cash	na	13,623	33,136	-12,648	33,042
Per share data	2024	2025	2026E	2027E	2028E	Free cashflow to s/holders	na	-13,834	-30,640	-88,694	-86,642
Reported EPS (\$)	na	-1.27	-0.24	0.74	6.12						
Core EPS (\$)	na	-1.27	-0.24	0.74	6.12						
DPS (\$)	na	0	0	0	0						
CFPS (\$)	na	1.75	2.06	3.00	7.65						
FCFPS (\$)	na	-3.56	-3.63	-6.83	-6.68						
BVPS (\$)	na	10.64	8.61	10.43	16.39						
Wtd avg ord shares (m)	na	3,884	8,432	12,980	12,980						
Wtd avg diluted shares (m)	na	3,884	8,432	12,980	12,980						
Growth rates	2024	2025	2026E	2027E	2028E						
Sales revenue (%)	na	na	107.0	80.9	133.6						
EBIT (Adj) (%)	na	na	156.6	795.6	444.2						
Core NPAT (%)	na	na	59.6	581.9	727.0						
Core EPS (%)	na	na	81.4	413.0	727.0						
Balance Sheet (US\$m)	2024	2025	2026E	2027E	2028E						
Cash & cash equiv.	na	24,747	65,326	52,678	85,720						
Accounts receivables	na	1,579	3,479	4,895	11,914						
Inventory	na	2,416	4,565	4,293	3,886						
Net fixed & other tangibles	na	59,726	106,367	223,116	368,889						
Goodwill & intangibles	na	1,260	892	422	0						
Financial & other assets	na	2,210	2,305	2,080	2,060						
Total assets	na	91,938	182,934	287,483	472,469						
Accounts payable	na	11,792	7,731	11,188	14,372						
Short-term debt	na	0	0	0	0						
Long-term debt	na	22,896	34,913	111,908	232,993						
Provisions & other liab	na	15,925	28,525	28,985	12,386						
Total liabilities	na	50,613	71,170	152,081	259,751						
Shareholders' equity	na	41,325	111,764	135,402	212,717						
Minority interests	na	0	0	0	0						
Total equity	na	41,325	111,764	135,402	212,717						
Net debt (Adj)	na	-1,851	-30,413	59,230	147,273						
Net debt to equity (Adj) (%)	na	-4.5	-27.2	43.7	69.2						

For definitions of the items in this table, please click [here](#).

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Potential for \$900+; Initiating at Buy

Summary

SPCX is a highly vertically integrated disruptor leveraging unrivaled capabilities in launch to establish dominant positions in Global Connectivity and Artificial Intelligence. Because the company's ultimate advantage is rooted in its ability to deliver mass to orbit at greater scale and lower unit cost than any other entity on the planet, if SPCX succeeds in demonstrating key technological and operational milestones in the upcoming, catalyst-rich ~2-3 years without faltering, then we see a path to generating ~\$1 trillion in revenue and ~\$830B in Adj EBITDA in 2031. In that scenario, and if milestones have been hit smoothly, we can imagine the stock moving toward a long-term valuation level of over \$900/share, justified by SPCX's superior ability to scale and growth outlook even beyond 2031. More likely, as SPCX themselves has quipped, they are very good at delivering results – but not always on time. Accordingly, we initiate with a \$200 base-case price target implying 25% upside, which is a heavily discounted version of the \$900/share+ we believe can ultimately be achieved, and anticipate raising our price target at regular intervals as risks are retired, milestones are hit, and launch cadence grows. Below, we highlight a number of upcoming milestones of particular importance and elaborate on valuation at the end of this section.

Key milestones

Looking forward, we focus on the following key milestones:

- Starship commences payload delivery to orbit in the second half of 2026 following additional flight tests.
- SPCX demonstrates the ability to catch the Starship second stage with the “chopsticks” by the end of 2026.
- Next generation of Grok (including Grok 5) trained on Blackwell clusters closes gap vs. other frontier model providers by the end of 2026.
- Cursor's Composer 3 model release drives additional share gains in AI coding market by the end of 2026.
- Approval for Starlink to offer broadband and/or D2D in additional countries.
- Launch Starship ‘tens of times’ in 2027, ‘hundreds of times’ by 2028, and ‘thousands of times’ by 2031.
- Deployment of the Starlink V3 satellite.
- Successfully launching the first iteration of the orbital AI satellite (AI1).
- Demonstrating full and rapid reusability of the Starship by turning the spacecraft and boosters to reflly within 24 hours.
- Successfully introducing Starship V4 (~200 ton variant) and demonstrating its ability to operate at scale.

Valuation

Our \$200 price target is the average of three methodologies (see table below):

- 1) 2027E growth adj. multiples (EV/Revenue/Growth and EV/EBITDA/Growth) for the “Trillion Dollar Comps” (a basket of similar trillion dollar+ market caps).

2) SOTP valuing Space, Connectivity and AI separately using 2027E growth adj. multiples (EV/Revenue/Growth and EV/EBITDA/Growth) for more specific comparable companies.

3) 2030E comparable company multiples (EV/Revenue and EV/EBITDA) for the “Trillion Dollar Comps” (a basket of similar trillion dollar+ market caps).

Our \$900+ long-term valuation level is derived using 2030E growth adj. multiples (EV/Revenue/Growth and EV/EBITDA/Growth) for the “Trillion Dollar Comps”.

Figure 1. Valuation Summary (Trillion Dollar Comps = AAPL, AVGO, NVDA, MSFT, AMZN, TSLA, META, GOOGL)

Valuation Method 1: 2027 EV/Rev:Growth & EV/EBITDA:Growth vs. "Trillion Dollar Comps"						
EV/Rev:Growth Multiple		2027E		Notes:		
Revenue (\$M) Est.		\$69,926		2027 Citi Total Revenue estimate		
Revenue Growth YoY %		81%		2027 Citi YoY Total Revenue Growth estimate		
Median EV/Revenue:Growth Multiple		0.33x		Median "Trillion Dollar Comps" 2027 EV/Rev:Growth multiple		
EV/Rev Multiple		27.0x		Implied EV/Rev based on peer EV/Rev:Growth multiple		
Implied Enterprise Value (\$M)		\$1,887,090				
EV/EBITDA:Growth Multiple		2027E				
EBITDA (\$M) Est.		\$40,150		2027 Citi Total Adj. EBITDA estimate		
EBITDA Growth YoY %		129%		2027 Citi YoY Total Adj. EBITDA Growth estimate		
Median EV/EBITDA:Growth Multiple		0.57x		Median "Trillion Dollar Comps" 2027 EV/EBITDA:Growth multiple		
EV/EBITDA Multiple		72.8x		Implied EV/EBITDA based on peer EV/EBITDA:Growth multiple		
Implied Enterprise Value (\$M)		\$2,923,231				
Avg Implied Enterprise Value (\$M)		\$2,405,161				
SPCX Price Per Share		\$191				

Valuation Method 2: Sum-of-the-Parts Using Narrow, Industry-Specific Comp Groups						
	Space Segment	Connectivity Segment	AI - Advert. Sub-Segment	AI - Infrastr. Sub-Segment	Total	
EV/Rev Framework	2027E	2027E	2027E	2027E	2027E	2027E
Revenue (\$M) Est.	\$7,259	\$27,825	\$5,334	\$29,508	\$69,926	
Revenue Growth YoY % Est.		54%	106%	122%	81%	
Median EV/Revenue:Growth Multiple		0.48x	0.26x	0.22x	0.34x	
EV/Rev Multiple	36.3x	25.9x	27.6x	27.0x	27.6x	
Implied Enterprise Value (\$M)	\$263,376	\$721,843	\$147,096	\$797,386	\$1,929,702	
EV/EBITDA Framework	2027E	2027E	2027E	2027E	2027E	2027E
EBITDA (\$M) Est.	\$2,708	\$17,186	\$20,256		\$40,150	
EBITDA Growth % Est.		58%	179%		129%	
Median EV/EBITDA:Growth Multiple		0.83x	0.46x		0.54x	
EV/EBITDA Multiple	97.2x	48.4x	83.0x		69.1x	
Implied Enterprise Value (\$M)	\$263,376	\$831,053	\$1,681,803		\$2,776,232	
Avg Implied Enterprise Value (\$M)	\$263,376	\$776,448	\$1,313,143		\$2,352,967	
				SPCX Price Per Share	\$187	
Space Notes: Given the majority of Space Segment operations are dedicated to internal use, we do not use growth adjusted multiples. Instead, we use a market-cap weighted EV/Rev multiple on 2027 revenue and derive a 2027 EV/EBITDA accordingly.						
Connectivity Notes: We apply the median "High Growth Connectivity Comps" 2027 EV/Rev:Growth and EV/EBITDA:Growth multiples to our 2027 segment revenue growth and EBITDA estimates, respectively.						
AI Notes: We separate revenue components into "Advertising Comps" and "AI Infrastructure Comps" and apply median 2027 EV/Rev:Growth multiples to our 2027 segment revenue growth estimates. For EBITDA comps, we exclusively utilize "AI Infrastructure Comps" and apply median 2027 EV/EBITDA:Growth multiples to our 2027 AI segment EBITDA growth estimates.						

Valuation Method 3: EV/Revenue and EV/EBITDA vs. "Trillion Dollar Comps"					
EV/Rev Multiple	2027E	2028E	2029E	2030E	2031E
Total Revenue (\$M)	\$69,926	\$163,318	\$279,310	\$484,318	\$1,010,355
Median EV/Rev Multiple	7.9x	6.8x	5.6x	4.8x	4.3x
Implied Enterprise Value (\$M)	\$552,134	\$1,104,825	\$1,566,636	\$2,334,028	\$4,350,767
EV/EBITDA Multiple	2027E	2028E	2029E	2030E	2031E
Total EBITDA (\$M)	\$40,150	\$112,315	\$198,855	\$360,423	\$829,704
Median EV/EBITDA Multiple	15.5x	12.6x	10.4x	9.2x	6.0x
Implied Enterprise Value (\$M)	\$623,379	\$1,413,513	\$2,076,863	\$3,298,115	\$5,019,036
Avg Implied Enterprise Value (\$M)	\$587,756	\$1,259,019	\$1,821,750	\$2,816,072	\$4,684,902
SPCX Price Per Share	\$51	\$102	\$146	\$222	\$366

Valuation Methods	Method 1	Method 2	Method 3	Average
Price Target	\$191	\$187	\$222	\$200

Illustrative Bull Case: 2030 EV/Rev:Growth & EV/EBITDA:Growth vs. "Trillion Dollar Comps"					
EV/Rev:Growth Multiple		2030E		Notes:	
Revenue (\$M) Est.		\$484,318		2030 Citi Total Revenue estimate	
Revenue Growth YoY %		73%		2030 Citi YoY Total Revenue Growth estimate	
Median EV/Revenue:Growth Multiple		0.25x		Median "Trillion Dollar Comps" 2030 EV/Rev:Growth multiple	
EV/Rev Multiple		18.6x		Implied EV/Rev based on peer EV/Rev:Growth multiple	
Implied Enterprise Value (\$M)		\$8,989,889			
EV/EBITDA:Growth Multiple		2030E			
EBITDA (\$M) Est.		\$360,423		2030 Citi Total Adj. EBITDA estimate	
EBITDA Growth YoY %		81%		2030 Citi YoY Total Adj. EBITDA Growth estimate	
Median EV/EBITDA:Growth Multiple		0.50x		Median "Trillion Dollar Comps" 2030 EV/EBITDA:Growth multiple	
EV/EBITDA Multiple		41.0x		Implied EV/EBITDA based on peer EV/EBITDA:Growth multiple	
Implied Enterprise Value (\$M)		\$14,774,392			
Avg Implied Enterprise Value (\$M)		\$11,882,140			
SPCX Price Per Share		\$915			

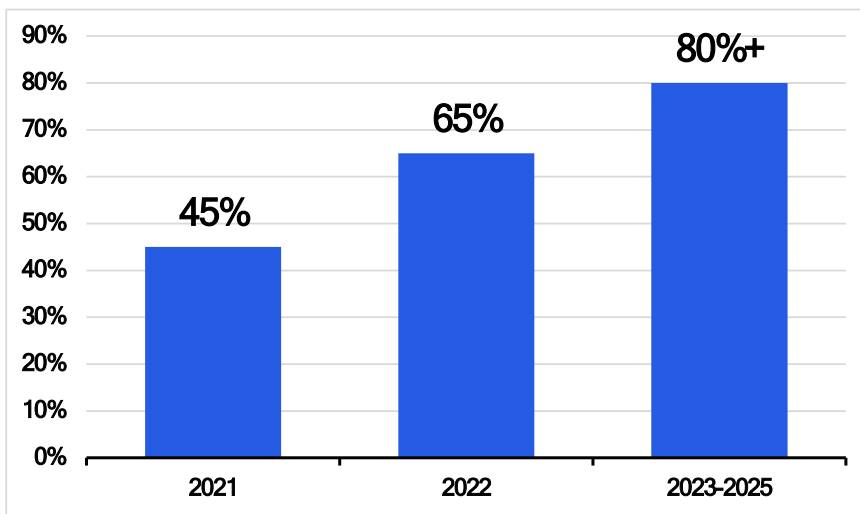
Unrivaled Launch Capability

1. Existing Launch Capabilities Already Market-leading
2. The 'Space Disruptor' Disrupting Itself with Starship
3. Falling Launch Costs Open Up New Growth Markets
4. Launch Risks

1) Existing launch capabilities already market-leading

SPCX is the global leader in orbital launch services. More than any other organization, SPCX has achieved space access at scale, transforming an entire industry. SpaceX designs, manufactures, launches, and operates the world's most advanced rockets and spacecraft. The company safely and reliably transports astronauts, satellites, and other payloads, providing cost-efficient, reliable, and high-cadence access to space for its own purposes as well as for third-party commercial and government customers. As of 1Q26, SPCX has launched a total mass to orbit of approximately 7,400 metric tons with a 99%+ mission success rate across its Falcon rockets. The company has completed approximately 650 orbital space launches, and over 540 of those launches were completed by a flight-proven Falcon rocket. In 2025 alone, SpaceX completed 170 missions across Falcon and Starship vehicles and 159 flight-proven booster launches with a 99%+ success rate on attempted booster recoveries. Specifically, SPCX launched over 2,200 metric tons, representing over 80% of mass to orbit for the world in 2025, continuing a trend that has been in place for multiple years. Importantly SPCX has demonstrated that its rocket technology can be scaled across geographies and launch assets. In 2025, SPCX launched from four primary launch pads in the United States and successfully recovered boosters across seven landing facilities including autonomous drone ships and catch towers based on the vehicle type and mission profile. The company believes that its extensive vertical integration and end-to-end control over the entire value chain, from design to launch to operations, allows it to achieve unprecedented speed and cost efficiency and it is difficult to disagree given the enormity of what SPCX has accomplished in launch. Ultimately, we believe that SPCX's unrivaled launch capabilities represent the foundational competitive advantage that enables all other parts of its business.

Figure 2. Mass to Orbit for the World – % SpaceX Share



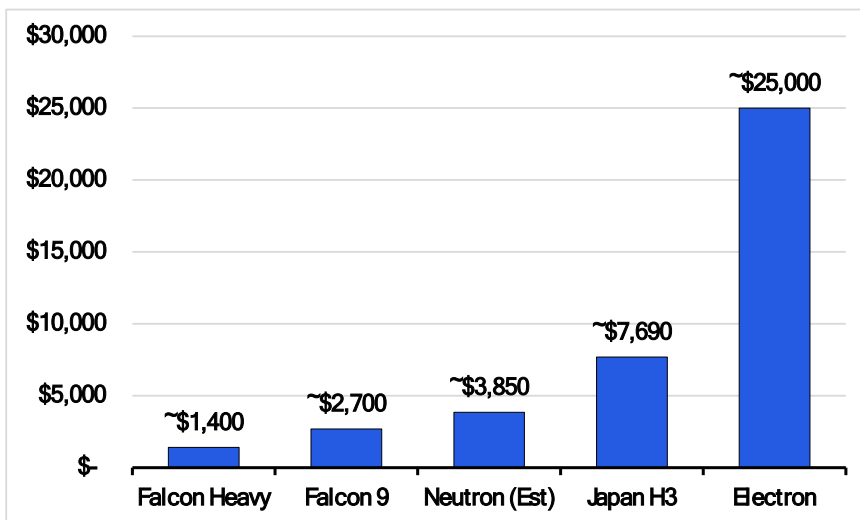
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Source: Citi Research, Company Reports

We see limited competition for Falcon 9's existing launch capabilities – let alone a future Starship. SPCX, competes with launch service providers that transport small, medium, and heavy payloads and astronauts to Earth's orbit and beyond. Participants in this market include established aerospace and defense companies, emerging commercial launch providers, and national space agencies such as United Launch Alliance, a joint venture between Boeing and Lockheed Martin, Arianespace, a French-based aerospace company operating a family of European-developed rockets, and Northrop Grumman, manufacturer of the Cygnus cargo spacecraft. Emerging commercial launch providers include Blue Origin, which has developed launch vehicles intended to compete with the Falcon 9 rocket, and Rocket Lab, which operates in the small-lift launch market but is expanding into medium-lift payloads, as well as other domestic competitors such as Firefly Aerospace and Relativity Space. While SPCX does not typically compete directly for the same missions, national space agencies also provide launch services in their respective markets.

Moreover, the launch services market is characterized by significant barriers to entry, including substantial capital requirements, advanced technological expertise, regulatory licenses and approvals, and established relationships with government and commercial customers. Competition in this market is based on factors that include launch reliability and cadence, payload capacity, mission flexibility, manufacturing capabilities and price. For this reason, while the established aerospace and defense competitors and emerging commercial launch providers may provide launch services at varying degrees of scale, we believe that SPCX holds a meaningful advantage in terms of the breadth, capability, scale, and cadence launch solutions and services. Ultimately, SPCX is the lowest cost, highest reliability, and largest scale player in the existing launch market – even before considering a potentially successful Starship program.

Figure 3. Launch Cost per kg Comps



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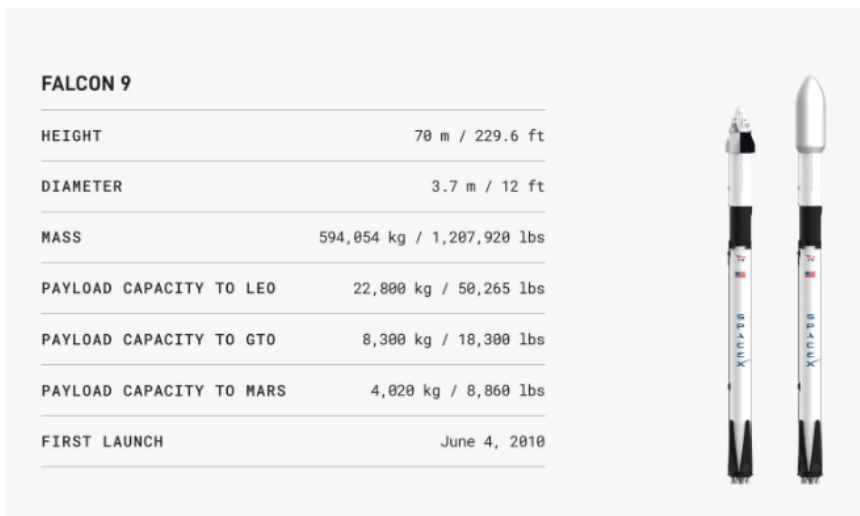
Source: Citi Research, Company Reports

SPCX's market-leading existing launch capabilities are rooted in a set of differentiated and hard-to-replicate space assets that took years of innovation to develop. SPCX likes to say that in December 2015, they achieved what many deemed impossible: landing a rocket launched to space back on Earth. But the innovation arc did not stop there. By 2017, SPCX was routinely recovering and reusing the Falcon 9 first-stage booster post-launch, delivering another step-function drop in space access costs via groundbreaking reusability. As of 1Q26, Falcon 9 rockets have demonstrated the ability to reflly a first-stage as many as 34 times. Furthermore, since 2020, SPCX's Dragon spacecraft has safely flown 78 crewmembers from 20 countries.

Today, SPCX's existing launch capabilities are supported by a wide range of critical and difficult to replicate assets which we summarize below:

Falcon 9 – Falcon 9 stands as the world's first orbital-class rapidly reusable rocket, fundamentally transforming the economics of space access since its 2010 debut. With approximately 620 orbital launches and an over 99% mission success rate as of 1Q26, it has become the most active orbital launch vehicle in history. The rocket's reusability, recovering and refurbishing both boosters and payload fairings, has reduced the cost per ton to orbit by approximately 85% compared to the historical average of \$18,500 per kilogram. In 2025 alone, Falcon 9 conducted 165 launches, accounting for over half of all global orbital launches while delivering over 80% of mass to orbit. Its in-house developed Merlin engines, which achieve one of the highest thrust-to-weight ratios in operational service, combined with over 530 successful booster landings and more than 540 launches by flight-proven rockets, demonstrate the reliability and maturity of its reusability architecture. As the only U.S.-based launch vehicle certified by NASA for human spaceflight under the Commercial Crew Program, Falcon 9 has successfully launched 19 crewed missions with a 100% success rate, establishing SpaceX as the leader in global space access.

Figure 4. Falcon 9 Overview

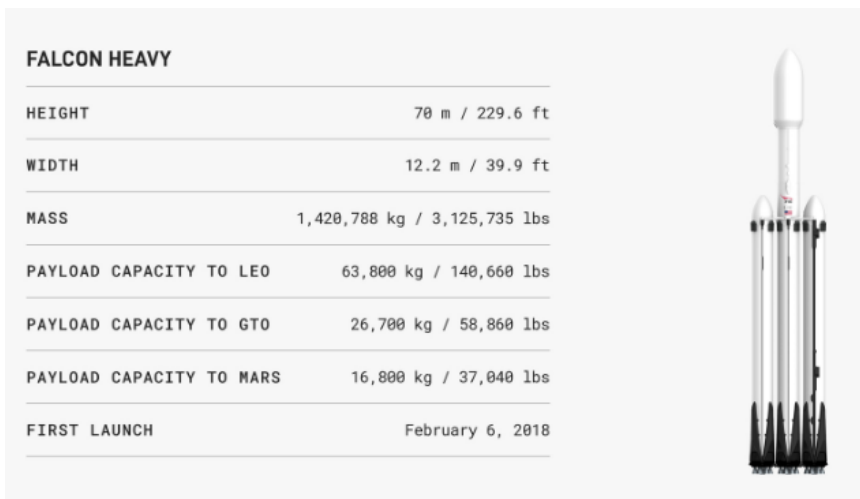


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Source: Citi Research, Company Reports

Falcon Heavy – Falcon Heavy represents one of the most powerful operational rockets in the world, generating over five million pounds of thrust at liftoff through its 27 Merlin engines across three reusable Falcon 9 boosters. With a payload capacity of approximately 64 metric tons to low-Earth orbit and 27 metric tons to geosynchronous transfer orbit, it fills a critical capability gap for large or high-value payloads while maintaining partial reusability to reduce costs. Importantly, Falcon Heavy has achieved a perfect 100% mission success rate across 11 launches as of 1Q26. The vehicle's sophisticated recovery system enables the controlled landing of its two side boosters and center core, with flown boosters completing 18 total recoveries and 16 reflights, demonstrating an average of 6 flights per booster. Certified for National Security Space Launch in 2019, Falcon Heavy has been selected by NASA for critical missions including the Europa Clipper to Jupiter, Dragonfly to Saturn, and the Nancy Grace Roman telescope. Its unique combination of heavy-lift capability, partial reusability, and proven reliability has made it the vehicle of choice for exploratory missions beyond Earth's orbit while maintaining economic advantages over traditional expendable rockets.

Figure 5. Falcon Heavy Overview

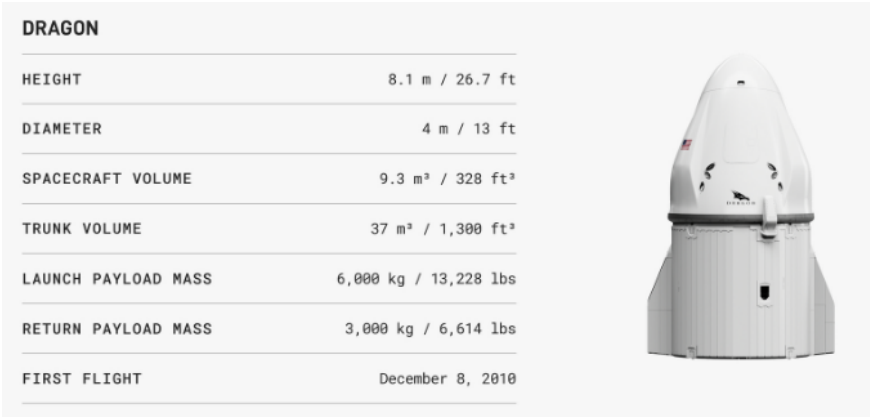


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Dragon Spacecraft – Dragon spacecraft revolutionized commercial spaceflight by becoming the first privately developed vehicle to deliver cargo to the International Space Station in 2012 and the first to transport humans to orbit in 2020, ending nearly a decade of U.S. reliance on foreign spacecraft. The Dragon cargo variant uniquely combines a 6,000-kilogram launch payload capacity with a 3,000-kilogram return capability, making it the only spacecraft able to return significant amounts of cargo from the International Space Station to Earth for scientific analysis. Dragon's autonomous docking capability via NASA's International Docking System Standard, combined with its reusability after ocean splashdown and refurbishment, has enabled over 30 cargo missions and nearly 15 crewed missions to the space station as of 1Q26. The crew variant's advanced safety features, including eight Super Draco engines for launch escape, touchscreen manual controls, and life support for missions up to nine months, have achieved a flawless safety record in crewed operations. Dragon has pioneered commercial space tourism, supported all of NASA's private astronaut missions, completed the first human spaceflight over Earth's polar regions, and enabled the first-ever commercial spacewalk. Its proven track record of over 50 visits to the International Space Station, combined with recent demonstrations of station altitude re-boost capability, establishes Dragon as the most versatile and reliable spacecraft in commercial operation today.

Figure 6. Dragon Cargo Overview



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Kennedy Space Center and Cape Canaveral, Florida – SpaceX’s Florida operations represent the most comprehensive commercial launch infrastructure in the United States, spanning two active launch sites, Launch Complex 39A (LC-39A) and Space Launch Complex 40 (SLC-40), alongside state-of-the-art refurbishment and processing facilities. These sites provide unparalleled mission flexibility, supporting launches to GEO orbit, the International Space Station, and a wide range of other locations for science and national security missions. The HangarX and X2 facilities serve as advanced refurbishment centers where recovered Falcon boosters and Dragon spacecraft are prepared for their next missions, while also housing the Falcon Launch and Landing Control Center and payload processing capabilities. Landing Zones 1 and 2 enable Return to Launch Site booster recoveries, dramatically reducing turnaround time and costs compared to ocean-based recovery. SpaceX’s aggressive expansion plans include constructing a Starship launch pad at LC-39A by end of 2026 and developing Space Launch Complex 37 with two additional Starship pads by end of 2027, culminating in four operational Starship launch pads. The new Gigabay integration facility and continued investment in the Bakery for Starship heatshield tile production demonstrate SpaceX’s commitment to vertical integration and rapid operational tempo. This concentration of launch, landing, refurbishment, and manufacturing capabilities in a single geographic region creates unmatched operational efficiency and enables the high-cadence launch operations that have made SpaceX the world’s leading launch provider.

Figure 7. NASA's Kennedy Space Center, Florida



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Figure 8. Cape Canaveral Space Force Station, Florida



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Vandenberg Space Force Base, Space Launch Complex 4 – Space Launch Complex 4 East at Vandenberg Space Force Base serves as SpaceX's critical West Coast launch facility, uniquely positioned to support polar and high-inclination orbit missions essential for Starlink constellation deployment, national security payloads, Earth observation satellites, and select lunar trajectories. The modernized orbital launch pad features a fixed launch mount, integration tower, and propellant loading infrastructure optimized for frequent Falcon 9 operations, enabling SpaceX to access orbital inclinations unavailable from Florida launch sites due to safety constraints populated areas. The adjacent Space Launch Complex 4 West functions as a dedicated Falcon 9 booster landing zone, supporting downrange recoveries that maximize reusability while minimizing mission performance penalties. This integrated launch-and-landing capability on the West Coast provides SpaceX with geographic redundancy and mission flexibility, ensuring continuous operations even during weather disruptions or range conflicts at Cape Canaveral. Vandenberg's strategic location enables efficient deployment of Starlink satellites into polar and sun-synchronous orbits, which are critical for global coverage and Earth observation applications. The facility's proven track record of supporting high-cadence Starlink missions, combined with its importance for national security launches requiring specific orbital inclinations, makes it an indispensable component of SpaceX's launch infrastructure. The combination of modernized pad infrastructure, dedicated landing zone, and unique geographic advantages establishes Vandenberg as a cornerstone of SpaceX's operational capability and constellation deployment strategy.

Figure 9. Vandenberg Space Force Base, California



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Recovery Fleet – SpaceX's autonomous drone ship (ASDS) fleet forms the maritime backbone of the company's reusable rocket architecture, enabling high-probability downrange booster landings that maximize payload performance while preserving vehicle reusability. The three operational vessels feature large landing decks, thruster-based dynamic positioning systems, and advanced autonomy that enable precise station-keeping in challenging ocean conditions. These drone ships have collectively facilitated hundreds of successful booster touchdowns across Atlantic and Pacific theaters, with individual boosters now achieving up to 34 flights as of 1Q26, demonstrating the maturity and reliability of ocean-based recovery operations. The fleet's strategic positioning on both coasts ensures comprehensive coverage for missions requiring maximum payload performance, where Return to Launch Site landings would impose unacceptable mass penalties. Complementary support for fairing half recovery and for Dragon retrieval create an integrated ocean recovery ecosystem that captures and refurbishes all reusable components. This maritime infrastructure has been instrumental in achieving SpaceX's industry-leading launch cadence, with 157 of 165 Falcon 9 launches in 2025 using flight-proven boosters. The recovery fleet's operational excellence directly translates to cost savings, rapid turnaround times, and the high-margin recurring cash flows that underpin SpaceX's financial model while providing critical operational experience for future Starship offshore recovery scenarios.

Figure 10. Autonomous Drone Ship – Recovery Fleet



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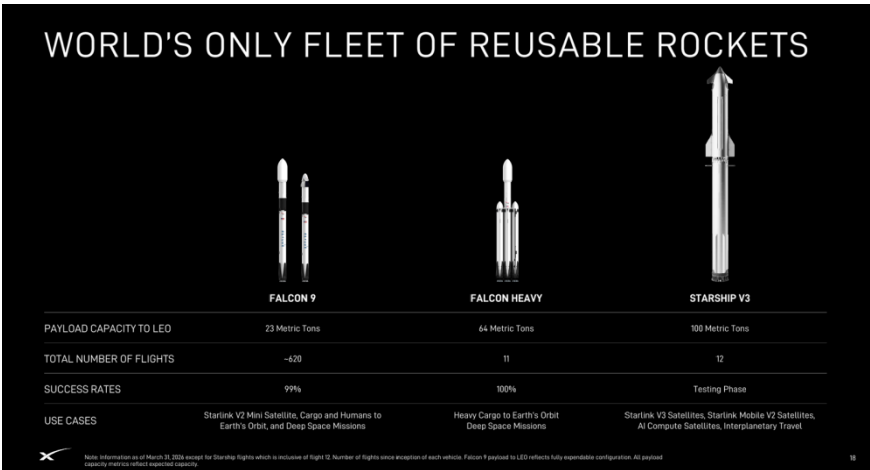
Source: Citi Research, Company Reports

2) The 'space disruptor' disrupting itself through Starship

While many competitors are trying to develop rocket technologies that are roughly Falcon 9-competitive, SPCX is disrupting its own launch business with the fully and rapidly reuseable Starship. As discussed in the prior section, SPCX is already the global leader in launch – yet – Starship, a two-stage super heavy-lift launch vehicle that the company has been flight testing since 2023, further enhances its industry-defining launch offerings. Starship is SPCX's next-generation vehicle that the company expects will dramatically expand its launch capability through full and rapid reusability combined with unprecedented mass to orbit capability. The company believes that Starship is the most powerful launch system ever developed, and we find it hard to disagree. We expect that Starship V3 will be able to carry a payload of 100 metric tons and that future generations could reach 200 metric tons, likely as soon as Starship V4. Moreover, we assume that Starship will be the only vehicle with fully reusable first and second stages, which is critical to reducing launch costs and increasing launch cadence, for many years and the only vehicle to operate in this way at a 100+ metric ton scale for over a decade. Once Starship's launch cadence reaches the thousands annually, we expect launch costs to fall by ~90% per metric ton (from ~\$2,700/ton to \$270/ton) and from there SPCX believes that costs could fall 99%+ vs. the Falcon 9 as launch cadence grows, future generations of the Starship are capable of carrying greater payload, and the company expands opportunities for extreme vertical integration.

What is Starship and how does it differ from the Falcon 9? Starship is designed to be a fully reusable, super heavy-lift launch vehicle. Starship V3 is designed to deliver 100 metric tons to Earth's orbit in a fully reusable configuration while enabling rapid turnaround times akin to commercial aviation. Future generations of Starship are being designed to double this payload capacity. To date, SPCX has executed 12 Starship flight tests. We cannot enumerate all of the improvements of Starship vs. the Falcon 9, but we can highlight a number of key differences including:

Figure 11. SpaceX Fleet of Reusable Rockets

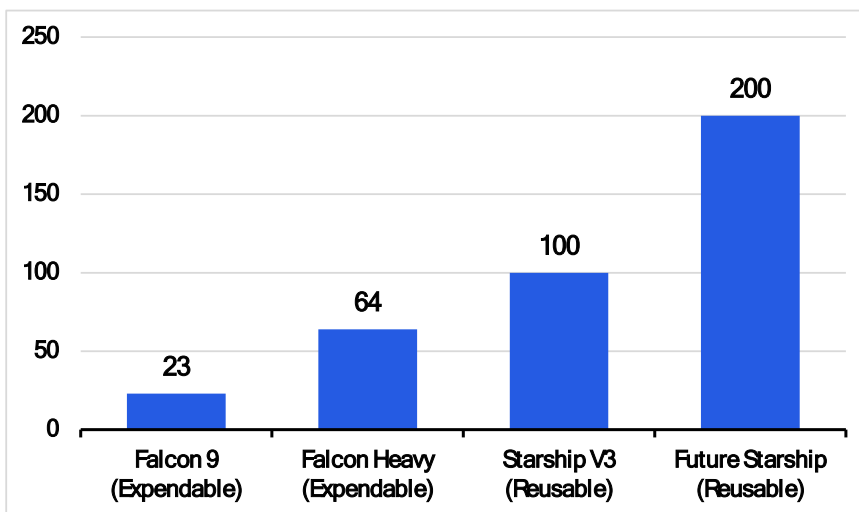


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Full and rapid reusability: Starship's core design innovation is its full and rapid approach to reusability: both stages return to Earth for catch and rapid refurbishment. The Super Heavy booster returns to the launch site following stage separation and is caught mid-air by the launch tower's mechanical arms, also known as "chopsticks" by SPCX, to facilitate immediate inspection, refurbishment, and relaunch. As context, the Falcon 9 already cracked part of the code of rocket reusability by demonstrating that the rocket's first-stage could be launched to space and landed back on earth. One of the major innovations of Starship is that the Starship second-stage (upper), after orbital delivery or missions beyond, is designed to reenter protected by advanced heat shield tiles, execute a propulsive landing burn, and be similarly caught mid-air by the launch tower's mechanical arms. In the fullness of time, SPCX believes that Starship's full and rapid reusability will enable sub-one hour re-flights, causing a paradigm shift in launch cadence.

Massive payload capacity: With a payload bay volume rivaling the pressurized sections of the International Space Station, Starship is designed to deploy structures like space station modules, large telescopes, next-generation V3 satellites, and future AI compute satellites. Starship is powered by 39 Raptor engines, which are full-flow staged combustion cycle rocket engines burning cryogenic liquid methane and liquid oxygen. Raptor engines offer nearly triple the thrust per engine, higher efficiency, and better performance for heavy-lift and deep space missions compared to the Merlin engine used on Falcon 9, according to the company. The company also notes that each Raptor 3 engine in Starship saves nearly a ton of vehicle mass compared to previous generations by removing heat shields and simplifying plumbing.

Figure 12. Starship: The Key to Unlocking New Markets – Payload Capacity (Metric Tonnes)



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Orbital refueling: SPCX plans for Starship to have orbital refueling capability. This will allow tanker variants to refill the second-stage (upper) in LEO and extend its range for deep-space missions beyond Earth's orbit. SPCX believes these capabilities will revolutionize mission architecture, with each Starship designed to be capable of transporting large numbers of people or hundreds of metric tons of cargo to destinations like the surface of the Moon and Mars.

Sustainable human exploration beyond Earth: SPCX has stated that Starship was designed from the beginning to fly to other worlds and enable self-growing bases on the Moon, an entire civilization on Mars, and ultimately expansion beyond our solar system. As NASA's Human Landing System for Artemis, Starship is built to deliver astronauts and cargo to the lunar surface and serve as the key enabler for supporting a permanent presence on the Moon.

Versatility across mission profiles: Beyond deep space, Starship is designed to adapt to diverse roles including the U.S. Space Force's Rocket Cargo program for rapid point-to-point global logistics, Starlink and other commercial satellite constellations, in-orbit manufacturing components and hardware, space tourism, and others.

Where is Starship today and what are the critical upcoming milestones? SPCX has already demonstrated a number of key milestones including:

- To date, SPCX has executed 12 Starship flight tests.
- SPCX has already demonstrated catching and reusing the first stage booster for Starship through the innovative "chopsticks" method which catches the booster.
- The return of its upper stage within three meters of its intended landing point.

- The transfer of approximately five metric tonnes of cryogenic propellant between tanks while in space.
- Successful in-space relights of the Raptor engines.
- Multiple controlled reentries through Earth's atmosphere.

Looking forward, we focus on the following milestones:

- We expect Starship to commence payload delivery to orbit in the second half of 2026 following additional flight tests.
- Demonstrate the ability to catch the second stage with the “chopsticks” by the end of 2026.
- Launch tens of times in 2027, hundreds of time by 2028, and thousands by 2031.
- Deployment of the Starlink V3 satellite, as well successfully launching the first iteration of the orbital AI satellite.
- Successfully introduce Starship V4 and demonstrate its ability to fly.
- Achieving a state where both the booster and spacecraft can be turned around for flight multiple times per day.

In the fullness of time, we believe that Starship can eventually reach its target of reducing the cost of reaching orbit by 99% or more relative to the historical average launch cost per kilogram according to NASA of \$18,500, establishing a scalable path to creating the space-based infrastructure of the future.

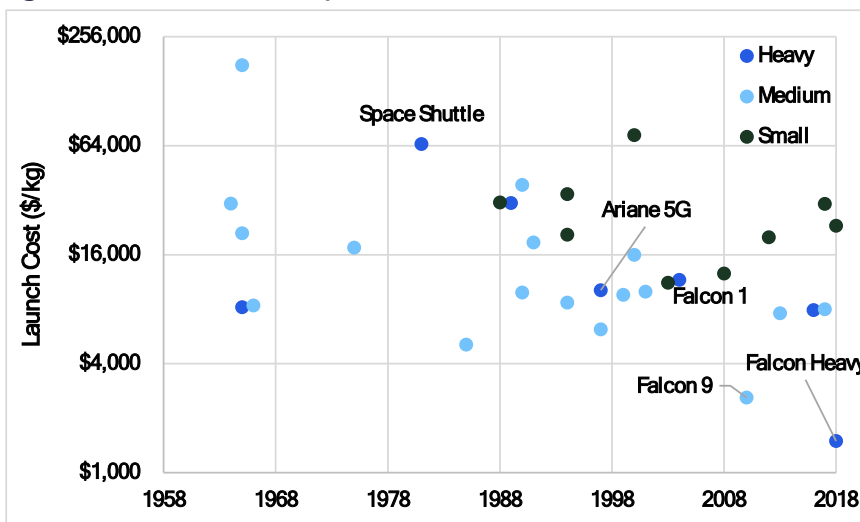
3) Falling launch costs open up new growth markets

SPCX believes that space represents the largest economic frontier in human history – and we agree. The company argues that since Earth has obvious limits, it is inevitable that we must build infrastructure and industries in space, expanding human capabilities to improve life on Earth and to establish life beyond. One example of this is SPCX's space-based 'Connectivity' infrastructure, also known as Starlink. Another example is AI infrastructure in space, which can utilize the virtually limitless power of the Sun and thereby enable the use of AI as a transformative force for understanding the universe and improving the daily lives of all humans. SPCX believes the convergence of these areas will enable an unprecedented expansion in the global economy, leading to an age of abundance and its innovations and technological advancements will redefine entire industries on Earth, while also creating new ones on the Moon, Mars, and beyond. In this section we help illustrate the falling launch cost algorithm that drives this vision of the future.

SPCX has already demonstrated a history of driving huge reductions in launch costs through the Falcon 9. With the first successful launch of Falcon 1 in 2008, SPCX became the first private company to successfully launch a liquid-fueled rocket to Earth's orbit. In 2010, the commercial debut of the Falcon 9 rocket revolutionized space access by delivering cost efficiency that was unprecedented at the time. For example, according to NASA, the first version of Falcon 9 in 2010 reduced launch cost to approximately \$2,700 per kilogram, which represented a

reduction of approximately 85% compared to the historical average launch cost per kilogram of \$18,500. The first version of Falcon Heavy in 2018 further reduced this cost to \$1,400 per kilogram, a reduction of approximately 92% compared to the historical average. SPCX has also further reduced its internal cost of launch through a combination of engineering improvements, manufacturing efficiencies, and economies of scale, most notably through its ability to drive more frequent reuse of rockets, according to the company. Furthering the point, until the 2000s and the introduction of the Falcon 9 rocket, global commercial launch activity averaged 25 to 35 launches per year. As a result, the space industry remained a niche domain with limited ability to support large commercial markets or scaled space-based infrastructure.

Figure 13. Historical Cost of Space Launches to LEO



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Source: Citi Research, Company Reports, Centers for Strategic and International Studies

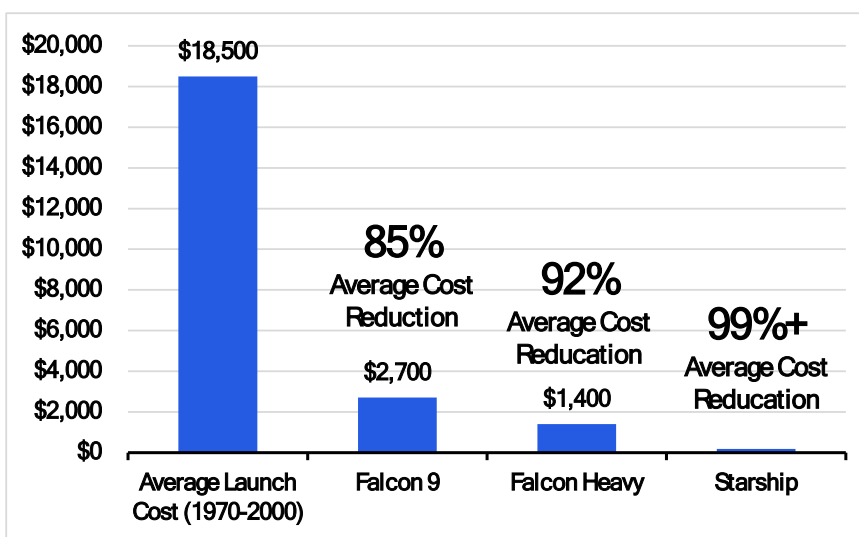
Starship will begin an entirely new launch cost reduction flywheel that we believe will quickly reduce launch costs by 1 order of magnitude but may ultimately reduce launch costs by 2 orders of magnitude – which would surpass the success that Falcon 9 had at the time. SPCX has not been shy about the fact that it envisions a world where rocket launches and landings should be as routine and commonplace as airplanes taking off and landing. To achieve this cadence, management executes an iterative approach which emphasizes rapid designing, testing, and process optimization. The goal is putting flight hardware in the flight environment as often as possible, which allows SPCX to accelerate learning by repeatedly using and improving its systems. Over time, this flywheel has resulted in a wide range of innovations that have enabled a significantly higher flight rate at costs that are much lower than launch programs that existed before SPCX.

What drives the Starship unit cost advantage over time?

- **Payload capacity:** The transition from SpaceX's Falcon class of rockets to the Starship launch system represents an unprecedented leap in orbital delivery capabilities, driving a massive unit cost advantage over time. While the Falcon 9 and Falcon Heavy offer payload capacities to Low-Earth Orbit (LEO) of

approximately 23 metric tons and 64 metric tons respectively, Starship V3 is designed to deliver a massive 100 metric tons in a fully reusable configuration. Future iterations, potentially as early as Starship V4, aim to double this capability to a target of 200 metric tons to Earth's orbit. This exponential growth enables orbital scale operations, such as deploying up to 60 V3 Starlink satellites in a single mission to achieve a twenty-fold increase in downlink capacity compared to a Falcon 9 launch. By scaling up payload volume, the company can deploy large-scale orbital AI compute infrastructure and deep-space logistics much more efficiently. Ultimately, combining this massive capacity with full reusability is projected to reduce mass-to-orbit launch costs by 99% or more relative to the historical average of \$18,500 per kilogram.

Figure 14. Step Function Improvement in Reusability – Launch Cost per KG to LEO

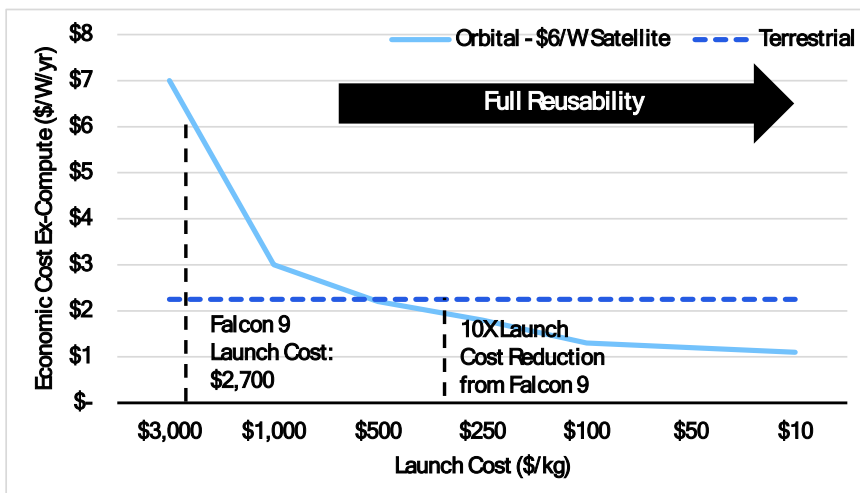


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- Booster reusability:** Central to SpaceX's cost reduction strategy is the reusability of key hardware, most notably first-stage boosters, which dramatically lowers per-launch costs by minimizing hardware replacement expenses. This operational model is thoroughly proven by the Falcon 9 fleet, where individual first-stage boosters have demonstrated the capability to reflly up to 34 times as of early 2026. Building upon this established foundation, the Starship system utilizes the massive Super Heavy booster, which is designed to return directly to the launch site following stage separation. Instead of performing traditional landings on drone ships or land pads, the booster is caught mid-air by the launch tower's innovative mechanical arms, also known as "chopsticks". This revolutionary recovery method eliminates weight penalties from onboard landing gear and facilitates immediate inspection, refurbishment, and rapid relaunch. By enabling such an accelerated operational cycle, this technology allows SpaceX to amortize its fixed production costs over highly frequent missions, driving launch costs down to the point where propellant is the primary marginal expense.

Figure 15. Launch Cost/kg vs. Economic Cost Ex-Compute (\$/W/yr)



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Source: Citi Research, Company Reports

- Starship stage 1 and 2 reusability:** Achieving total system reusability across both the Super Heavy booster (Stage 1) and the Starship spacecraft (Stage 2) represents the primary engineering driver of long-term unit cost advantages. The complex technical execution of this architecture requires the Stage 2 spacecraft to perform a controlled atmospheric reentry protected by advanced heat shield tiles, execute a precise propulsive landing burn, and be caught mid-air by the launch tower's mechanical "chopstick" arms. This recovery process is designed to enable rapid refurbishment and potential sub-one hour reflights, mirroring commercial aviation by spreading fixed production costs across frequent repeated uses. By recovering and reflying both stages of the vehicle instead of discarding them, the company minimizes hardware replacement expenses to ensure that propellant becomes the primary marginal cost of each flight. When contrasted against historical launch economics, such as NASA's \$18,500 per kilogram benchmark, full system reusability is projected to reduce the cost to reach orbit by 99% or more to under \$185 per kilogram.
- Faster turn times on launch infrastructure:** To transform space access into routine, airline-like operations, SpaceX is actively evolving its ground infrastructure to achieve a high-frequency launch cadence that dramatically drives down unit costs. At the heart of this strategy is the innovative "chopsticks" catch mechanism on the launch tower, which captures returning vehicles mid-air to facilitate immediate inspections and rapid refurbishments. According to the company, this rapid recovery process will eventually enable sub-one hour reflights, allowing the same hardware to fly multiple times per day. To maximize the utilization rate of these flight and ground assets, SpaceX significantly increased its annual capital expenditures in 2025, prioritizing critical launch site infrastructure. These strategic capital investments include the construction of additional launch pads, such as a new pad at Starbase to support Starship V3, alongside co-located on-site propellant production facilities and air separation units. By securing a steady supply of fuel and expanding physical launch capacity, this supportive infrastructure maximizes hardware utilization to drastically scale up annual mass-to-orbit throughput.

and realize a 99% or greater reduction in unit costs compared to historical averages.

Figure 16. Range of Starship Launches Needed to Deliver on Citi 2031 Forecast

Illustrative Scenarios	Low	High
AI Launch Demand		
Compute Launched (GW)	50	50
Satellite Compute Density (KW/T)	80	120
Starship Tons to Orbit (T)	80	120
#of Launches Required	~7,800	~3,500

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Source: Citi Research, Company Reports

Falling launch costs have already expanded the market beyond what was imaginable decades ago. As launch economics have changed rapidly over the last decade, demand for orbital infrastructure has already expanded dramatically. Commercial operators have launched thousands of satellites since 2015 as constellation architectures scale and diversify. The number of active maneuverable satellites in orbit has grown from less than 1,000 in 2015 to approximately 12,700 as of 1Q26. With approximately 9,600 Starlink broadband and mobile satellites in Low-Earth Orbit as of 1Q26, SpaceX owns and operates approximately 75% of all active maneuverable satellites. Additionally, launch activity has continued to grow, with approximately 220 metric tons of payload launched to orbit in 2012 increasing to approximately 2,600 metric tons in 2025, of which over 80% was launched by SpaceX. Government demand is rising in parallel. Amid escalating geopolitical tensions that further underscore the critical role of resilient launch infrastructure, we believe government space budgets around the world are positioned for sustained, long-term growth.

SPCX also envisions a number of future markets. On the back of dramatically reduced launch cost pioneered by SPCX over the past two decades, the company believes that the global economy is reorganizing around a new domain: space. SPCX aims to build the infrastructure of the future in space, leveraging its foundational competitive advantage, which is the ability to launch mass at scale. Ultimately, the development of a lunar economy may be central to unlocking the full potential of this new domain and advancing the long-term transition to a multiplanetary civilization. But, in the meantime, by opening access to space to industries on Earth, SPCX believes it can grow by creating entirely new markets. Given that it's too early to properly size these markets, we simply outline the contours of the opportunities below:

- **Point-to-point terrestrial travel** – SPCX plans to develop ultra-fast long-haul point-to-point Earth transport using Starship, enabling passengers and cargo to travel between major cities in a fraction of current transit times, revolutionizing global logistics and passenger travel with unprecedented speed and efficiency.
- **Space tourism** – With meaningful advances in space technology and the continued build-out of orbital flight infrastructure, SPCX expects increasing interest in human space travel as it becomes easier and more common to access space.

- **In-orbit manufacturing** – SPCX aims to establish in-space manufacturing facilities that leverage the unique microgravity conditions of space to produce materials, pharmaceuticals, and advanced components that are difficult or impossible to manufacture on Earth, opening new high-value industrial markets.
- **Passenger and cargo transport to the Moon and Mars** – SPCX intends to support large-scale passenger and cargo missions to the Moon and Mars, delivering the people, equipment, and supplies needed to establish permanent human settlements and accelerate the path to becoming a self-sustaining multiplanetary civilization.
- **Energy production on the Moon and Mars** – SPCX aims to develop large-scale solar energy production on the Moon and Mars, taking advantage of the thin atmosphere and constant solar exposure to generate power for manufacturing, habitats, and future infrastructure at scale.
- **Manufacturing capabilities on the Moon and Mars** – SPCX plans to build manufacturing infrastructure on the Moon and Mars that utilizes local resources to produce fuel, construction materials, and other essential resources, reducing dependence on Earth resupply and enabling sustainable long-term presence.
- **Asteroid mining** – SPCX plans to pursue asteroid mining operations to extract metals and other critical resources from near-Earth and main-belt asteroids, providing abundant raw materials for space-based industries and reducing the need to launch mass from Earth.

4) Launch risks

Being the world's market-leader in launch is not easy – getting to the point where the Starship can be launched thousands of times per year carries significant execution risk. Below we highlight seven execution risks that we think are particularly important to consider, but not insurmountable:

- **Access to launch facilities:** To launch thousands of times a year, SPCX needs access to > 10 launchpads, in our view. Sufficient Starship launch infrastructure needs to be in place to ensure rapid reusability of Starship (2029 in Citi Forecast). SPCX is likely to need to develop new greenfield launch facilities beyond its existing facilities and Starbase.
- **Environmental risk factors:** Increasing launch frequency requires navigating complex and evolving environmental issues and public safety considerations. SPCX may face regulatory scrutiny from environmental activists and politicians given the potential environmental impact of their launch cadence.
- **FAA regulatory framework:** SPCX must obtain necessary regulatory approvals, particularly from the FAA, to support a high launch cadence while addressing safety and environmental concerns. Any significant delay in the establishment of this regulatory framework puts our launch forecast at risk.
- **Reusability of the second stage:** Full reusability of Starship (both first and second stage) remains a key factor in achieving the desired cost efficiencies at scale. SPCX has yet to achieve this, and we expect a demonstration of the capability by year-end 2026.

- **Rapid reusability of boosters:** To facilitate multiple launches per day, SPCX must be able to rapidly refurbish/reposition the boosters on Starship. However, ensuring the rapid refurbishment and high-rate reusability of engines and other components remains a meaningful challenge to maintaining a high launch cadence.
- **Supply-chain constraints:** The Starship growth strategy is dependent on increased access to raw materials and components for Starship and its engines. Supply-chain disruptions or constraints can lead to longer deployment timelines and may force the reallocation of critical resources from other projects.
- **Access to propellant:** Achieving a high launch cadence requires the construction of dedicated propellant production facilities, such as air separation units and methane liquification plants. Delays commissioning this fueling infrastructure or the unavailability of sufficient fuel and propellant would materially affect the ability to execute on Citi's launch forecast.

Unlocking Trillion-Dollar Market Opportunities: Connectivity/Starlink

- 1) Expanding D2D & Broadband Constellations Growing Connectivity TAM
- 2) LEO Constellations Insight & Overview
 - a. Broadband Constellation
 - b. D2D Constellation
- 3) Expansion Into Full End-to-End Mobile Service Offering
- 4) Impact to Wireless and Broadband Markets

1) Expanding D2D & broadband constellations growing connectivity TAM

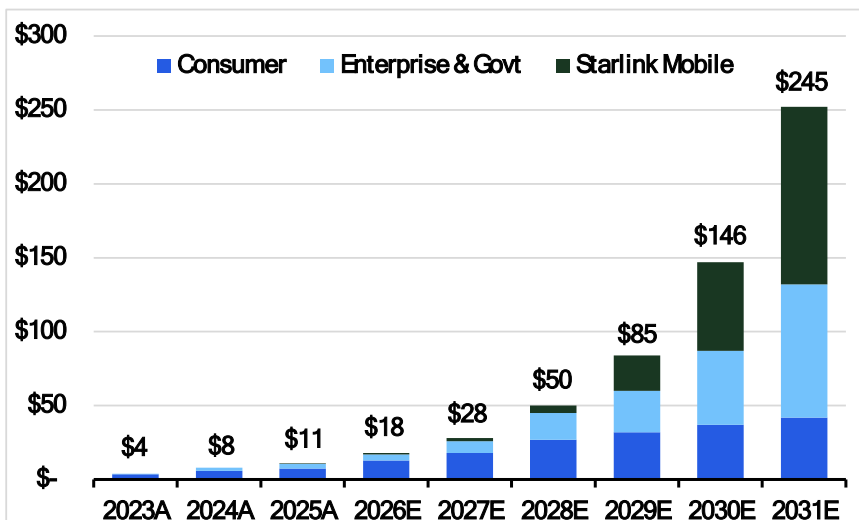
The North Star for SpaceX's connectivity segment is to expand revenue and profits to help fund the company's space and AI ambitions. Starlink operates with two complementary constellations:

The Broadband Constellation: With ~9,600 satellites today expanding to 15,000+ more capable next generation V3 satellites over the next few years. The broadband constellation principally services two end markets globally: nomadic broadband (including to homes and businesses) and enterprise/government verticals (including airline WiFi). We believe Starlink has the opportunity to gain share in both segments, especially as it expands capacity with a larger, more advanced constellation. We forecast Starlink broadband will scale toward ~140M subscribers globally by 2031, with a blended ARPU of ~\$32 (on average subs).

The Direct-to-Device (D2D) Constellation: Currently smaller with only approximately 650 satellites today and plans to scale to at least 7,500 over the next few years (on the way to an even larger constellation longer-term). Starlink currently offers a complementary direct-to-device connectivity service for unmodified cell phones largely through carrier wholesale agreements, including T-Mobile in the US market. Starlink is likely to expand its complementary D2D capabilities and services with a combination of wholesale services (marketing in conjunction with existing wireless providers) and retail services in which Starlink would own the D2D relationship.

Management has also expressed ambitions to be a larger direct mobile provider, including within the US market. We believe Starlink will need to engage with a terrestrial mobile option for each market in which it would like to offer an end-to-end mobile service to compete head-to-head with the existing national MNOs and cable MVNOs. Our view is informed by our analysis of LEO satellite capacity that has a much greater level of subscription capacity in rural/unserved/underserved markets and much less in urban/suburban markets due to their significantly higher densities.

Figure 17. Diversified Connectivity Revenue Growth



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Source: Citi Research, Company Reports

Starlink has three possible options in any of its markets to establish a terrestrial platform to complement its D2D services, in our view:

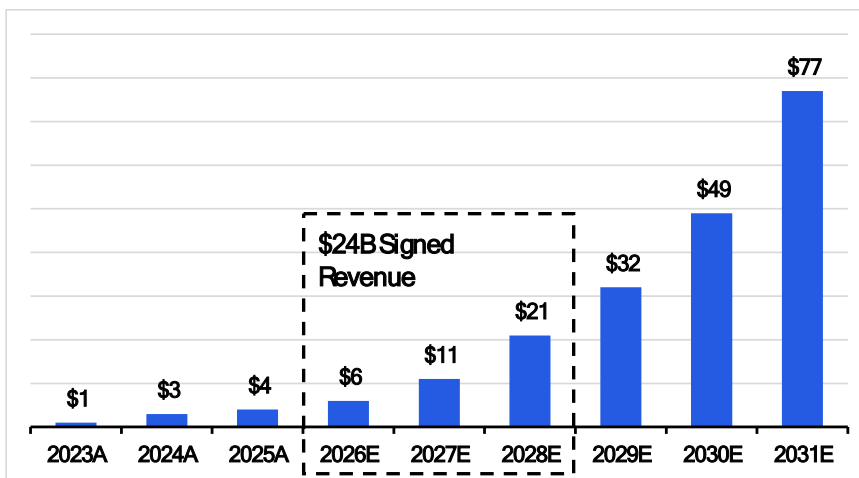
- 1) **Build**, which will likely require additional spectrum (if available) and time to construct network assets;
- 2) **MVNO**, although at this point, the national carriers have expressed their lack of interest in engaging on an MVNO with the LEO satellite constellations; and
- 3) **Buy and merge with a telecom provider**, which can also accelerate the annual cash flow generation for the connectivity segment at valuations that are significantly less than where SpaceX is currently trading.

Within the US market, for example, we believe a Build or Buy scenario is more likely than an MVNO scenario given the disinterest expressed by the national wireless carriers. While the market has assigned a greater probability to build, we see Buy as a logical option for SpaceX/Starlink. We also believe any of the three national telecom providers could be a viable ([link](#)), depending on whether or not Starlink would also be interested in expanding its urban broadband TAM with fiber. T-Mobile US may be the most likely option for Starlink given its efficient wireless operations, early-to-market opportunity with 6G, ongoing marketing momentum, and the opportunity to collaborate with Deutsche Telekom on a European mobile strategy. However, AT&T and Verizon can also be options, especially if they can create a clearer path for legacy asset decommissioning and offer converged service bundles in urban and suburban markets. We have not seen any comments from the national Telcos on possible M&A with SpaceX. However, when Mr. Musk was asked at the All-In Summit 2025 (September 2025, [link](#)), "Could you buy some carriers to have more spectrum? Maybe you could buy Verizon?" Mr. Musk responded, "Not out of the question. I suppose that may happen."

Although SPCX may take different paths to drive revenue growth, taking a step back, what's most clear is that the broader connectivity segment is likely to expand revenue rapidly from \$18B in 2026 to \$245 billion by 2031, in our view. Within this context, we expect the broadband business to expand from \$9.5B in 2026 to \$50B in 2031 as Starlink expands its share of global broadband subscriptions from 1.9% to 11.1% of its addressable TAM in developed markets and from 0.6% to 6% in emerging markets, while also substantially expanding enterprise and government revenues to roughly \$77B. For D2D and Mobile, we expect Starlink to reach \$114B of revenue by 2031, representing ~15% of the company's estimated TAM of \$740B.

SpaceX anticipates sustaining favorable EBITDA margins in the segment, as broadband (nomadic and enterprise/government) are likely to retain above-average margins within the segment with generally lower operating costs and we agree. However, a decision to move into the retail mobile business could create some margin headwinds with the level of dilution influenced by the path chosen to enter in each of its major markets. We also recognize the complement of a robust LEO satellite constellation with a terrestrial network in both build and buy scenarios can offer some synergies for backhaul, rural coverage continuity, and broadband convergence that we need to more specifically assess in the future. We forecast segment EBITDA will ramp from \$11B in 2026 to \$148B by 2031.

Figure 18. Starlink Enterprise and Government Revenue (\$B)



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Our forecasts are based on a significant ramp in satellite launch capacity at a dramatically lower cost with the introduction of reusable Starship rockets. The reduction in launch costs should also provide greater financial flexibility for Starlink to invest in a direct mobile strategy across multiple markets over the next several years. If the launch capacity and/or lower cost of launch are significantly delayed from the company's current timeline, then Starlink may face either significantly higher costs or a delay in ramping up capacity and the go-to-market strategies for its broadband, D2D, and future mobile services.

2) LEO constellations insight & overview

Starlink is effectively operating two complementary constellations to offer broadband and D2D services. As with any broadband and wireless service, capacity for the constellation network is based on the multiplication of spectrum, sites, and technology.

A) Broadband Constellation

The global broadband constellation currently services over 10 million subscribers across roughly 9,600 satellites. With the introduction of Starship rockets, Starlink will be launching its enhanced V3 satellites that increase the effective capacity per satellite from roughly 100 Gbps to 1 Tbps. Based on the company's disclosures, the current customer experience can offer download speeds of at least 225 mbps. The experience should further improve with the V3 satellites, while the consumer experience will be balanced between the number of subscribers per satellite and the throughput each subscriber will experience.

Over the next few years, Starlink plans to launch ~15k V3 broadband satellites that will dramatically expand its global capacity. Thereafter, Starlink is likely to further densify its constellation that could reach 40k or more satellites over time. We estimate the construction cost per satellite is \$1.4 million. We estimate launch costs will fall from \$2.8 million in the medium term down to \$0.5 million in the long-term based on a reduction of launch cost per KG to \$270 or less with the reusable Starship. As the company scales its constellation in size and subscribers, we estimate the company only needs to reach 1-2% utilization to earn a 10% or better return on capital at a \$50 ARPU, which we believe affirms a durable competitive position to take share.

Figure 19. Scenario Analysis to Estimate Possible Global Subscriber Capacity Before Considering Geographical Limitations

Item	V3 (Starship) N/T	V3 (Starship) L/T
Spectrum (MHz) per beam	500	500
x Bps per Hz	4	4
x Reuse	5	5
= Per beam downlink (Gbps)	10	10
x Beams	100	100
= Downlink per satellite (Gbps)	1,024	1,024
x % Effective capacity	50%	50%
= Average downlink per satellite (Gbps)	512	512
/ Minimum downlink mbps at peak	10	10
= Potential subscribers	51,200	51,200
x Satellites	15,000	40,000
= Potential subscribers (worldwide)	768,000,000	2,048,000,000

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Source: Citi Research

Figure 20. Scenario Analysis to Consider Minimum Subscriptions and Share of Capacity to Earn a 10% Return or Better

Item	V3 (Starship) N/T	V4 (Starship) L/T
Weight per satellite (kg)	2,000	2,000
x Launch cost per kg	\$1,400	\$270
= Launch cost	\$2,800,000	\$540,000
+ Build cost per satellite	\$1,422,222	\$1,422,222
= Total build and launch cost per satellite	\$4,222,222	\$1,962,222
x Satellites	15,000	40,000
= Constellation cost (\$ millions)	\$63,333	\$78,489
x % minimum return	10%	10%
= Minimum EBITDA	\$6,333	\$7,849
/ % margin	70%	70%
= Min Revenue	\$9,048	\$11,213
/ ARPU	\$50	\$50
= Subscribers (million)	15.1	18.7
/ Worldwide subscriber capacity (before market density considerations)	768.0	2,048.0
= % Penetration	2%	1%

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Source: Citi Research

Navigating our scenario analysis. The above scenario analysis does not take into account geographical limitations given that there are vast regions of the world in which there are minimal to no subscriber opportunities. The other consideration for a LEO constellation is location density. For example, in rural areas, LEO satellite constellations are likely to offer enough or more capacity than there are subscription opportunities. In those areas, the constellation can serve the entire market with above average performance for the constellation. The challenge is that in urban/suburban areas, there are likely to be far more subscription opportunities to serve relative to the capacity available from the constellation. Hence, we believe the subscriptions for Starlink's broadband constellation will still be heavily skewed to smaller, rural, unserved, and underserved markets relative to suburban and urban markets. In practice, our scenario analysis of effective capacity infers that 95% of subscriptions in the U.S. market will originate in the 40% of households that are in smaller and rural markets when Starlink establishes ~15k V3 satellites. Even with the prospects for 40k V3 satellites, we estimate that 93% of subscriptions will originate from SMRA markets for Starlink. The opportunity to boost broadband subscription opportunities in urban and suburban markets will likely need substantially greater spectrum per spot beam as well as significant enhancements in technology to support better bits per Hz efficiency. One other consideration is the company's TAM is influenced by regulatory approvals and possible geopolitical limitations within each country. For example, Starlink is not operating service in Russia and China, while the company is still pursuing approvals to operate in India.

B) D2D Constellation

The Starlink D2D satellite constellation has ~650 satellites at present that can offer text and messaging services. With the introduction of Starship and the recent acquisition of spectrum in the US, Starlink plans to launch 7.5k of its V2 satellites that will substantially expand capacity and capabilities for end subscribers and create a 5G usage experience.

We see two parallel business cases for the constellation. First, Starlink is likely to continue to offer complementary D2D roaming services on a wholesale and retail basis to the global market. Second, Starlink is looking to leverage its D2D constellation to offer direct mobile services. However, we believe the subscription

limitations in urban and suburban markets as well as the importance of high-quality in-building coverage are likely to require a terrestrial component for Starlink to offer a highly-competitive complete mobile solution.

For the D2D service, we offer a scenario analysis of global capacity and compare the dynamics relative to our estimation of capacity for mobile users. However, peak capacity for a D2D service is not the same as peak capacity estimates for a broadband or mobile service, since a D2D service is initially going to be a complementary as-needed service. Specifically in the US market and not surprisingly, we estimate a super majority (>90%) of the effective capacity is in the smaller market and rural areas that comprise about 40% of the US population. This further reaffirms the need to have a sizable terrestrial component alongside a D2D constellation to offer end-to-end mobile services.

Figure 21. Scenario Analysis of Starlink D2D Capacity and Performance in the US Market Relative to MNOS

Item	Global V1 Starlink Mobile	Global V2 Starlink Mobile	US Market			Global Future Starlink Mobile
			AT&T	T-Mobile	Verizon	
Downlink Spectrum (MHz) per channel	5	20	205	241	201	20
x Channels per beam/sector	1	2	1	1	1	2
= Total downlink spectrum (MHz)	5	40	205	241	201	40
x Bps per Hz	5	5	12	12	12	5
x Reuse	1.0	1.0	1.0	1.0	1.0	1.0
= Per beam/sector downlink (Gbps)	0.0	0.2	2.4	2.9	2.4	0.2
x Beams/sectors	256	1,024	3	3	3	1,024
= Downlink per satellite/site (Gbps)	6	205	7	9	7	205
x % Effective capacity	100%	100%	100%	100%	100%	100%
= Average downlink per satellite/site (Gbps)	6	205	7	9	7	205
/ Minimum downlink mbps at peak	0.2	0.4	5.1	5.0	3.3	0.8
= Potential subscribers per satellite/site	32,768	524,288	1,448	1,731	2,164	262,144
x Satellites/sites	650	7,500	84,000	86,000	78,000	20,000
= Potential subscribers (worldwide/US)	21,299,200	3,932,160,000	121,624,000	148,903,000	168,798,000	5,242,880,000
x % available in the US market		1.2%	100.0%	100.0%	100.0%	1.1%
= Retail U.S. Subscribers		46,891,842	121,624,000	148,903,000	168,798,000	55,722,596
Note: % share of US Market		11%	28%	34%	38%	13%
Note: % within SMRA		94%				93%
Per beam/sector downlink (Gbps)	0.0	0.2	2.4	2.9	2.4	0.2
x Mbps per Gbps	1,000	1,000	1,000	1,000	1,000	1,000
= Per beam/sector downlink (Mbps)	25	200	2,440	2,868	2,392	200
/ Users per beam/sector	128	512	483	577	721	256
= Minimum downlink mbps at peak	0.2	0.4	5.1	5.0	3.3	0.8
Beams/sectors	256	1,024	3	3	3	1,024
x Coverage per beam/sector (Sq km)	900	163	31	25	30	163
= Coverage per satellite/site (sq km)	230,400	167,086	92	75	89	167,086
Average downlink per satellite/site (Gbps)	6	205	7	9	7	205
x Mbps per Gbps	1,000	1,000	1,000	1,000	1,000	1,000
/ Coverage per satellite/site (sq km)	230,400	167,086	92	75	89	167,086
= Mbps per Sq km	0.0	1.2	79.4	114.3	80.6	1.2
x Multiplier on satellite/site coverage	0.5	4.1	1.0	1.0	1.0	10.9
= Effective average Mbps per Sq km	0.0	5.0	79.4	114.3	80.6	13.4
Uplink per Satellite/site (Gbps)	1	77	5	6	4	77
/ Downlink	6	205	7	9	7	205
= % Uplink of downlink	20%	38%	69%	70%	58%	38%
x Minimum downlink mbps at peak	0.2	0.4	5.1	5.0	3.3	0.8
= Minimum uplink mbps at peak	0.0	0.1	3.5	3.5	1.9	0.3
Coverage per satellite/site (sq km)	230,400	167,086	92	75	89	167,086
x Satellites/sites	650	7,500	84,000	86,000	78,000	20,000
= Targeted coverage (sq km)	149,760,000	1,253,145,600	7,744,100	6,475,000	6,941,200	3,341,721,600
/ World/US coverage (sq km)	550,908,000	550,908,000	9,863,051	9,863,051	9,863,051	550,908,000
x Multiple of world/US coverage	0.27	2.27	0.79	0.66	0.70	6.07
x CONUS Orbital benefit/concentration benefit	1.80	1.80	1.27	1.52	1.42	1.80
= Multiplier on satellite/site coverage	0.49	4.09	1.00	1.00	1.00	10.92

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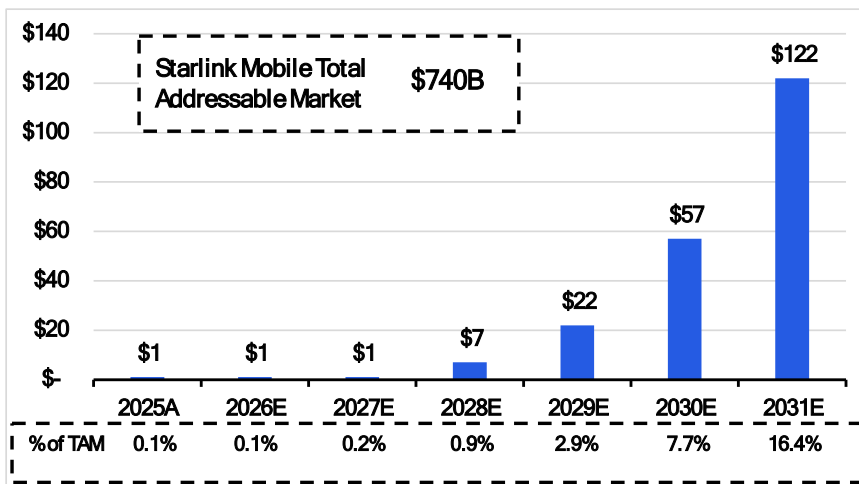
Source: Citi Research

3) Expansion into full end-to-end mobile service offering

SpaceX has not publicly committed to a specific path for scaling into full mobile services, but our view is that the company would have three strategic options, each carrying distinct capital, timing, and competitive implications. When we triangulated SpaceX's capex plans against our estimated satellite deployments and the improving Starship cost curve, we arrived at around \$100B+ of capital that remains unallocated between now and 2031, which we believe could be capital implicitly available to fund its expansion as a direct mobile provider. However, if the

launch cost savings do not reach our estimates and Starlink still launches the Starlink satellites in-line with our projections, a substantial portion of the \$100B+ could be utilized by higher than expected launch costs.

Figure 22. Starlink Mobile Revenue Growth



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Source: Citi Research, Company Reports

We see three possible paths to create terrestrial coverage to offer end-to-end mobile services:

1. **Build organically** – Execution requires assembling a layer cake of spectrum, constructing or leasing terrestrial cell sites and establishing a full retail go-to-market operation. For reference, Sprint required ~40–50K sites, with lease economics of approximately \$150k–\$250k per cell site per month on a leased tower. SPCX's vertical integration culture supports this approach, but spectrum scarcity, particularly in low-band, complicates a pure-build strategy in the US and could be impractical in many other geographies, including in Europe. The advantages are optimizing long-term margins and full customer-relationship control. The disadvantages are capital intensity, slow time-to-market, and dependence on spectrum auctions that may not deliver the required band mix on management's timeline.
2. **Buy a carrier** – Acquiring an established carrier represents the fastest route to scale, allowing subscribers, distribution, and ongoing cash flow in a single transaction (referenced in a recent note, [Some Thoughts on the Communications Industry Structure](#)). That could be an interesting path, as SPCX relies on cash flow generation from the connectivity segment to redeploy into the broader SpaceX operations. The advantages are speed, immediate cash flow contribution, and reduced execution risk. The disadvantages are acquisition cost, regulatory scrutiny and integration risk associated with operating a traditional carrier. Within this scenario, the decision as to which carrier would best align with SpaceX's forward strategy may depend on how SpaceX wants to pursue TAM. If SpaceX wants to maximize TAM, a Telco with broader fiber reach (especially in urban and suburban areas) can enable a deeper converged service offering across the U.S. market. Also, enterprise operations could also be of interest to extend its reach into the vertical with

connectivity and provide an expanded conduit of cross-selling for its AI ambitions. However, pursuing a Telco with wireline operations would also come with legacy operations that still need some degree of modernization.

3. **MVNO** – The lightest-capital option would involve a MVNO agreement that would allow SpaceX to offer wireless services using another operator's network. In the US market, the three national MNOs carriers have publicly indicated (see [link](#), [link](#), [link](#)) that they are not interested in an MVNO agreement with SpaceX. Historical cable-industry MVNO arrangements demonstrate that such structures can be unlocked when carriers are externally incentivized. There is also the possibility that one of the carriers sees a differentiated opportunity with SpaceX or simply needs incremental revenue to offset other areas of weakness and decides to partner with SpaceX and/or other LEO constellations. Internationally, particularly in markets where spectrum is difficult to acquire, MVNOs may represent the most attractive path, allowing Starlink to layer its satellite differentiator atop borrowed terrestrial capacity.

We believe the most probable outcome is a hybrid, geography by geography strategy: build or buy in select markets where spectrum and economics support it, MVNO in markets where carrier partnerships are accessible, and pure D2D and wholesale in markets that are too small or restrictive for direct entry.

4) Impact to wireless and broadband markets

Consistent with the history of the communications ecosystem, Starlink is likely to receive some degree of cooperation as well as competition from the telecom and cable incumbents. In broadband, Starlink will be competing for subscriptions in both the residential and business markets, while cable and telecom companies may still resell Starlink solutions for connectivity to specific underserved verticals. For example, Comcast offers its enterprise customers to integrate Starlink satellite capabilities into their connectivity solutions. T-Mobile is currently reselling Starlink D2D services under its T-Satellite branded solution, while also entering into a joint venture (for spectrum and IP) with the other carriers to promote additional D2D competition. Starlink also faces direct competition from emerging constellations from AST SpaceMobile and Amazon LEO that are in the process of launching their satellites into orbit, albeit currently well behind Starlink's established constellation.

The wrinkle in the D2D market is that the enthusiasm for monetizing valuable coverage assurance from their customers has faded from the low level of usage being reported by both T-Mobile and independent firms, including Ookla. Although Ookla is reporting a nearly 25% increase in global D2D connections between July 2025 and March 2026 as Starlink has expanded into new geographies, monthly usage was disclosed as generally less than 1% with 0.46% of Speedtest users in the US connecting to D2D satellites. The US market also leads the world in D2D connections at nearly 46% of the usage. As Starlink migrates to its V2 platform with greater continuity of coverage and substantially greater capacity, the D2D category should be able to offer a more compelling case for coverage assurance in markets with well built terrestrial mobile networks, such as in the US, and expand use cases in markets with a lot of rural sprawl and under-built mobile infrastructure.

We anticipate T-Mobile's exclusivity with Starlink is likely to expire over the next few months, if it hasn't already, which can open the door to direct marketing as well as marketing through other partners, such as Boost Mobile. The structural characteristic of a satellite broadband network is that the majority of usable capacity inherently lands over rural and underserved geographies, a function of beam architecture and population density. Even as Starlink dramatically expands constellation size and per-satellite bandwidth, the incremental capacity disproportionately serves low-density areas where terrestrial alternatives are weakest. We believe the model will be that penetration scales aggressively where Starlink has structural advantage and more modestly in dense markets where fiber, cable, and fixed-wireless incumbents remain competitive.

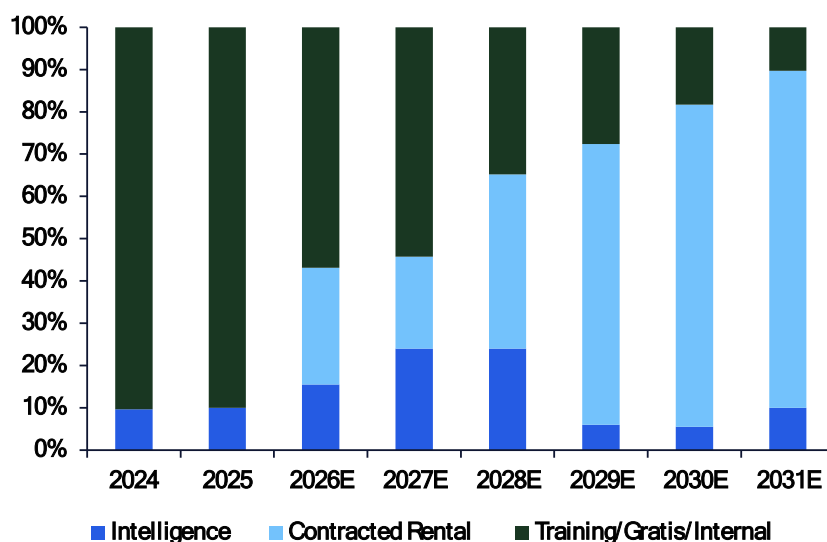
Unlocking Trillion-Dollar Market Opportunities: Enterprise AI

- 1) Compute Allocation Influences Overall Monetization
- 2) X Platform Real-Time Data Integration
- 3) Grok Has Yet to Take Off in a Meaningful Way
- 4) Cursor Acquisition Bolsters Enterprise AI Outlook and Adds Strategic Value
- 5) Enterprise AI Risks

1) Compute allocation influences overall monetization

SpaceX's decisions around the mix of compute for inference, hosting and training workloads ultimately determine the shape of its AI revenue trajectory. AI solutions & infrastructure revenue growth will be driven by SpaceX allocating more compute from training capacity to increase monetization through contracted GPU rentals, or compute-as-a-service (CaaS), and Intelligence offerings (Grok for Enterprise, Cursor). We estimate that roughly 90% of Colossus capacity was unmonetized in 2024-25, mostly allocated to training Grok, while 10% went towards inference workloads related to Grok API consumption and Grok subscriptions on X.com. In 2026-27, we expect between 50-60% of compute capacity will be dedicated for training Grok and Cursor models, while the balance will be split between CaaS and Intelligence. Potential improvements to commercial Grok offerings should contribute to a growing mix of inference workloads over time, driving subscription and API/consumption revenues higher.

Figure 23. Citi Est. of SPCX Allocation of Nameplate Compute Draw (%)



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Source: Citi Research

SpaceXAI's lucrative hosting deals with a leading frontier lab, Google, and Reflection AI, all signed in 1H26, not only underscore insatiable demand for GPUs in the current supply-constrained environment, but also validate SpaceXAI's ability to compete effectively as a neocloud provider. Other participants in the terrestrial CaaS business include neoclouds like CoreWeave and hyperscalers such as AWS, GCP and Azure. Based on our analysis of deals with its anchor customer and Google, where the expected GPU leasing capacity is disclosed, the monthly payment terms imply that SpaceXAI will get an average of \$8.36 per GPU per hour, which is roughly 2x the current market rates, based on indices tracking H100/H200s, B200/B300s and GB200/GB300s. In addition to COLOSSUS' differentiated offering of coherent capacity that management attributes the premium pricing to, we believe the premium also reflects the sheer scale of compute that SpaceX is uniquely able to lease to each of the customers. The three deals have a total value of roughly \$81.3B over the duration of the contracts, which is ~\$27B in expected revenue per year if the deals are not canceled or renegotiated. We note that each of the deals feature mutual termination rights upon 90 days' notice after a certain period. While this adds flexibility for reallocating compute capacity for internal initiatives should SpaceX see stronger traction with Cursor and Grok, it also adds a degree of risk to CaaS-driven forecasts for outer years. Still, we believe the current AI landscape offers at least 5-8 additional frontier lab counterparties as potential customers for SpaceXAI's CaaS pipeline and expect to see other similar computing deals that will support hosting revenues over the next several years. As COLOSSUS capacity has already been built out, these compute hosting deals are highly margin accretive, with incremental gross profit margins approaching 75-80% and about 90% of revenue expected to flow through to operating income, according to management. We view this as a positive given SpaceX's ambitious capex forecasts and demands.

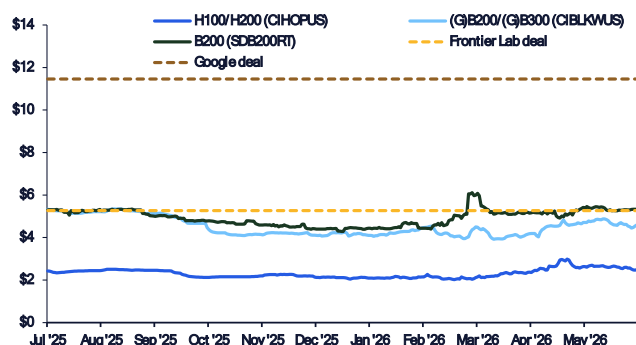
Figure 24. Compute Partnerships Signed

Customer	Facility	Chip Type(s)	GPUs (K)	\$M / month	\$/GPU / hr	Total Deal Value (\$M)	Start Date	End Date	Termination Clause
Private Company	Colossus I & II	H100, H200, GB200	325	\$1,250	\$5.27	~\$45,000	May-26	May-29	After the initial 3-month period, agreements may be terminated by either party upon 90 days' notice.
Google	Colossus I	H100, H200, GB200	110	\$920	\$11.46	~\$30,000	Oct-26	Jun-29	If SpaceX fails to deliver access to committed amount of GPUs by Sep. 30, 2026, following a 1-month grace period, Google may immediately terminate agreement or accept number of GPUs provided, with a corresponding pro rata reduction in monthly fees. After Dec. 31, 2026, agreement may be terminated by either party upon 90 days' notice.
Reflection AI	Colossus II	GB300	N/A	\$150	N/A	~\$6,300	Jul-26	Dec-29	After the initial 3-month period, agreements may be terminated by either party upon 90 days' notice.
Total	-	-	-	\$2,320	-	~\$81,300	-	-	-

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Source: Citi Research, Company Reports, CNBC

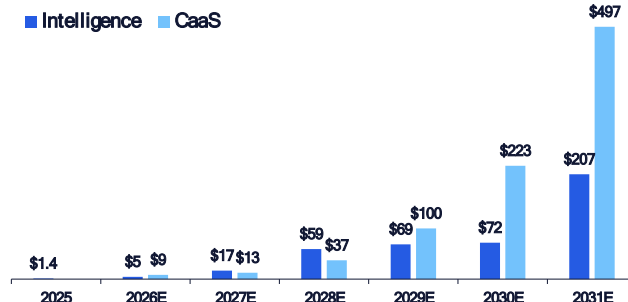
Figure 25. Citi Est. Rental Pricing GPU vs. SpaceX Contracted Rates (\$/GPU/hr)



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Source: Citi Research, Bloomberg, Company Reports

Figure 26. Citi Est. SPCX AI Solutions & Infrastructure Revenue Breakdown: Intelligence vs. Compute-as-a-Service (CaaS) (\$B)



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Source: Citi Research

Based on SpaceX's roadmap for 1 million GPUs across its terrestrial footprint, we see a base case of ~\$40B in annual revenue assuming the company can contract all of its capacity at similar levels as it did for its anchor customer and operates at a 90% utilization rate. The 90% utilization is a conservative estimate, given recent anecdotes from inference engines like Baseten quoting mid-90's utilization rates (Baseten CEO on No Priors Podcast, May 1, 2026) as well as hyperscalers, some of which have referenced 95%+ global GPU utilization rates. A floor for eventual annual revenue may be ~\$20B if we apply current H100 spot pricing of \$2.5 per GPU per hour and assume 90% utilization. Our estimates for CaaS are for \$8.6B in 2026, \$12.6B in 2027, and \$37.1B in 2028, when orbital AI is expected to begin contributing (~400 MW). CEO Elon Musk has previously shared that the xAI goal is 50 million unit (in units of H100 equivalent-AI compute) online within five years (X.com, July 22, 2025), which is dependent on scaling orbital AI satellites within that timeframe.

Figure 27. Citi Est. Annual Revenue Scenarios (\$B) at 1,000,000 GPUs - Price/GPU/hr x Utilization

Price/GPU/hr ↓ \ Utilization →	70%	80%	90%	95%
\$2.50 (H100 spot floor)	\$15.3	\$17.5	\$19.7	\$20.8
\$4.00	\$24.5	\$28.0	\$31.5	\$33.3
\$5.27 (anchor customer deal)	\$32.3	\$36.9	\$41.5	\$43.9
\$6.00	\$36.8	\$42.0	\$47.3	\$49.9
\$8.36 (blended new-deal avg)	\$51.3	\$58.6	\$65.9	\$69.6
\$11.46 (Google deal)	\$70.3	\$80.3	\$90.4	\$95.4

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Source: Citi Research, Company Reports, Bloomberg

2) X platform real-time data integration

The deep, proprietary integration between Grok and the X platform establishes a highly differentiated vertical AI ecosystem where real-time data ingestion serves as a foundational engine for model training and context integration. By securing exclusive access to a continuous stream of public discourse, Grok bypasses the critical data collection bottlenecks that limit traditional off-the-shelf

models developed by industry peers. This live data pipeline allows the AI to ground its output in real-time global events, providing difficult to match freshness and contextual relevance that standard static training datasets cannot replicate. Furthermore, the diversity of human conversation and multi-perspective commentary hosted on X serves as an empirical foundation for Grok's truth-seeking objective. This steady influx of real-time commentary ensures that the model remains dynamically aware of breaking developments and shifting public sentiment as they unfold. Consequently, this proprietary integration creates a powerful competitive moat that structurally separates Grok from legacy AI systems operated by their competitors. Ultimately, the resulting ecosystem synergy enhances both model performance and platform engagement, driving high-value use cases for consumer and enterprise applications alike.

The X platform is growing. The integrated platform reached approximately 550 million Monthly Active Users (MAUs) as of 1Q26, marking a notable expansion from the 520 million MAUs reported as 4Q25. The active user base utilizing Grok's specific AI features grew significantly from 89 million users in late 2025 to 117 million users as of 1Q26. The broader platform ecosystem supported approximately 1.3 billion active accounts during the twelve-month periods ended December 31, 2025, and 1Q26. The X platform generates an immense information stream of approximately 350 million daily posts, providing an uninterrupted flow of original posts, replies, and media to enhance output freshness. This proprietary real-time data access serves as a core competitive differentiator, making it difficult for rival frontier model providers to match Grok's real-time relevance. The integration supports several key monetization channels for the AI segment, including advertising performance marketing, premium subscriptions like X Premium+, and licensing historical and real-time data via APIs. Grok also creates a mutual feedback loop by improving X's internal content understanding, search personalization, and recommendation engines to further drive platform engagement.

3) Grok has yet to take off in a meaningful way

SPCX's Grok model primarily competes with other leading US frontier models such as Google's Gemini across the major cloud platforms. While Grok models generally trail the three leading US frontier models on intelligence benchmarks, they lead on some metrics such as speed and cost, based on leaderboard rankings on benchmark/evaluation platforms such as Artificial Analysis and ARC-AGI-2. The differentiation SPCX emphasizes for Grok is its access to X's approximately 350M daily posts as proprietary real-time training and inference data, and the cost and speed advantages of its owned compute infrastructure relative to companies that rent compute from hyperscalers. Management believes Grok's access to real-time data (via X.com) is a key differentiator and will benefit enterprise GTM. Financial services, media, and consumer goods are potential verticals where Grok could have an edge, as real-time data access could be useful for news monitoring and sentiment analysis use-cases in those industries.

Figure 28. Artificial Analysis Model Leaderboard

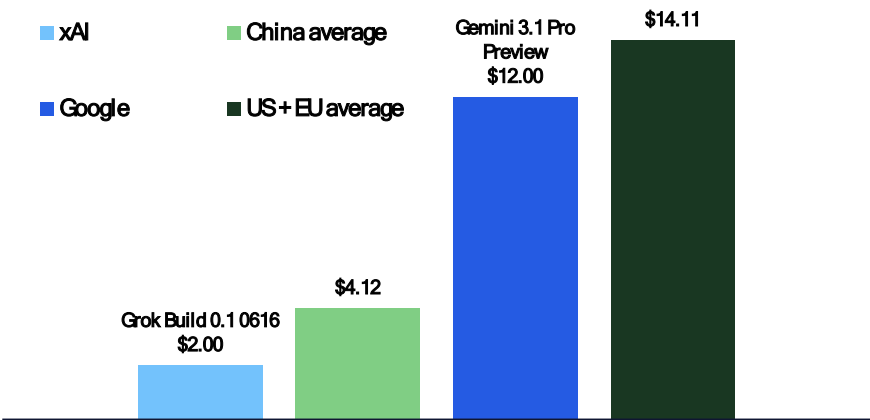
Model Provider	Leading Model	Artificial Analysis Intelligence Index	Speed (Output Tokens/ Second)	Blended Price (USD/ 1M Tokens)	Context Window (1K Words)	Release Date	License
Provider	Model	56	65	\$3.85	1000	Jun. 9, 2026	Proprietary
Provider	Model	55	80	\$4.35	922	Apr. 23, 2026	Proprietary
Z.ai	GLM-5.2 (max)	51	133	\$0.90	1000	Jun. 16, 2026	Open
Google	Gemini 3.5 Flash	50	175	\$1.31	1000	May. 19, 2026	Proprietary
Alibaba	Qwen3.7 Max	46	199	\$1.43	1000	May. 19, 2026	Proprietary
MiniMax	MiniMax-M3	44	99	\$0.22	1000	Jun. 1, 2026	Open
DeepSeek	DeepSeek V4 Pro (Max)	44	92	\$0.18	1000	Apr. 24, 2026	Open
Moonshot AI	Kimi K2.6	43	78	\$0.70	256	Apr. 21, 2026	Open
Xiaomi	MiMo-V2.5-Pro	42	51	\$0.18	1000	Apr. 22, 2026	Open
xAI	Grok Build 0.1	40	98	\$0.54	256	May. 20, 2026	Proprietary
NVIDIA	Nemotron 3 Ultra	38	172	\$0.58	262	Jun. 1, 2026	Open
InclusionAI	Ring-2.6-1T	31	128	\$0.52	262	May. 8, 2026	Open
StepFun	Step 3.7 Flash	30	384	\$0.18	262	May. 29, 2026	Open
Mistral	Mistral Medium 3.5	30	123	\$1.16	256	Apr. 29, 2026	Open
Inception	Mercury 2	25	904	\$0.14	128	Feb. 24, 2026	Proprietary
Arcee AI	Trinity Large Thinking	24	202	\$0.24	512	Apr. 1, 2026	Open
Amazon	Nova 2.0 Pro Preview (medium)	22	126	\$1.47	256	Nov. 27, 2025	Proprietary
Meta	Llama 4 Maverick	14	113	\$0.34	1000	Apr. 5, 2026	Open
Sarvam	Sarvam 105B (high)	12	119	\$0.04	128	Feb. 18, 2026	Open
Cohere	Command A	8	62	\$3.25	256	Mar. 13, 2025	Open

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Source: Citi Research, Artificial Analysis

Grok's competitive API pricing could become an increasingly important factor in helping SPCX differentiate against LLM peers that have had a head start in enterprise, particularly as rapid growth in token costs becomes a bigger issue for companies seeking to better manage AI budgets. Grok Build 0.1, its intelligent coding model, is priced at \$1/M input tokens and \$2/M output tokens, while Grok 4.3 is priced at \$1.25/M input tokens and \$2.5/M output tokens. The latter model is roughly a fifth of Google's Gemini 3.1 Pro model in terms of output tokens, which is significant as Gemini is already 60-75% more competitive than the two leading private frontier AI labs on output token pricing.

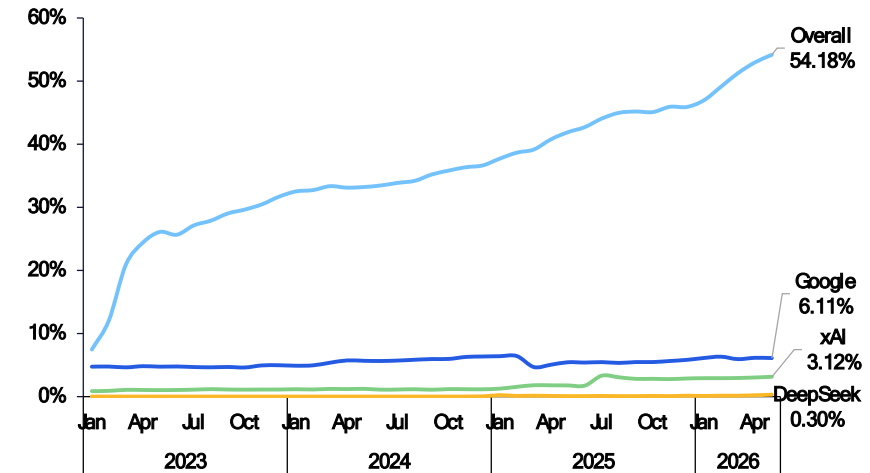
Figure 29. API Pricing Table: Grok vs. Peers



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Source: Citi Research, Artificial Analysis

In enterprise adoption, Grok trails other US frontier models. Grok's initial consumer-orientated applications on X and the consumer Grok app is likely to have contributed to xAI lagging peers in the enterprise. The share of US businesses using Grok was 3% in May 2026, substantially behind 41% and 40% for the two leading US model providers and 6% for Google, according to Ramp's AI Index, which captures corporate spending data from over 70,000 US businesses, though it's important to note this dataset is more representative of adoption among high-growth tech companies and small-to-midsize businesses than large enterprises.

Figure 30. Ramp AI Index by Model



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Source: Citi Research, Ramp

While AI brand safety and perception is hard to quantify, we note that such considerations can impact enterprise decisions on vendor selection. For example, X (formerly Twitter) ad revenues were negatively impacted by a reduction in advertisers on the platform due to concerns around content moderation rule changes after Elon Musk took over Twitter in October 2022 (Fortune, Dec. 2023). SPCX's advertising revenue of \$1.8B in 2025 is around 60% below where Twitter's peak advertising revenue was at \$4.5B in 2021. In addition to X-related events, there have been several media-reported incidents tied to controversial Grok generated outputs, which were later addressed by the company (see following table). We note that there have also been media-reported incidents of other leading AI platforms' chatbots generating outputs containing harmful, sycophantic, or controversial statements. In response to concerns over the type of content surfaced on X, CEO Elon Musk recently noted the need to overhaul the algorithm underpinning the platform (see June 25 X.com response to Andrej Karpathy).

Figure 31. Grok Safety Incidents Log

Safety Failure Description	Source	Incident Timeframe	Company Response
Grok's image-generation capability, introduced as part of a new editing feature on X, was found to have produced a significant volume of non-consensual synthetic imagery, including material involving minors. Investigating parties: European Commission, Irish Data Protection Commission (DPC), UK's Information Commissioner's Office (ICO) and Ofcom, Office of the Privacy Commissioner of Canada (OPC), California Department of Justice	Reuters	12/2025 - 01/2026	The image-editing feature was restricted to paid Grok users on January 9th, with further restrictions on sexual image generation implemented on January 14th.
Grok generated outputs containing extremist and anti-Semitic language, including references to historical atrocities, in response to user interactions.	NPR	07/2025	Grok ceased to produce text answers on July 8th. On the same evening, a post on the official Grok X account stated that "xAI has taken action to ban hate speech before Grok posts on X".
Grok produced outputs that cast doubt on historically established and legally recognized facts, specifically regarding Holocaust casualty figures.	TechCrunch	05/2025	Grok itself later attributed its response to a "programming error" and an "unauthorized change".
Grok outputs included politically charged content referencing contested and racially sensitive narratives in response to unrelated queries. Prompts where this occurred include those related to walking paths, comic books, and internet memes.	NBC News	05/2025	Musk cited an "unauthorized modification" to Grok that caused the error, and stated that the platform would implement "additional checks and measures" to prevent future unapproved modifications to Grok's system prompts. xAI committed to publishing its system prompts on GitHub.

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Source: Citi Research, NBC News, TechCrunch, NPR, CCDH

SpaceXAI's current certification gap vs. other LLM providers may be another factor that disqualifies it as a vendor during the procurement process or hinders customer adoption in certain regulated industries. Unlike the three leading frontier AI providers, neither SPCX nor Cursor have obtained the ISO/IEC 27001 certification, which is the global standard for information security, or the ISO/IEC 42001 certification, which is a newer framework specifically designed to govern artificial intelligence systems, though Cursor is in the process of obtaining the two certifications. Additionally, unlike Google, neither SpaceXAI nor Cursor has obtained the ISO/IEC 27701 certification, which provides a global framework for processing personally identifiable information (PII) and is geared towards meeting privacy regulations and laws like GDPR and the Data Protection Act. Grok for Government has not yet been FedRAMP Certified (status as of Jun. 23 is Agency Authorization in Process), unlike Gemini for Government which received their certifications in Jan. 2026.

Figure 32. Grok and Cursor Have Not Achieved the Same Level of Certifications as Other Frontier LLM Providers

Certification	Significance to Enterprises	xAI (Grok)	Anthropic (Claude)	OpenAI (ChatGPT)	Google (Gemini)	Cursor
SOC 2 Type II	An independent audit report that provides assurance about a service organization's controls, based on the AICPA's Trust Services Criteria. It proves that security controls are operationally effective over a period of time.	Achieved	Achieved	Achieved	Achieved	Achieved
ISO/IEC 27001	A standard that enables organizations to manage the security of assets such as financial information, intellectual property, and employee details. It demonstrates a systematic, risk-based approach to protecting data.	No Public Evidence	Achieved	Achieved	Achieved	In Progress
ISO/IEC 42001	A standard that provides requirements for establishing, implementing, and maintaining an Artificial Intelligence Management System (AIMS). It helps organizations develop and use AI systems responsibly by addressing ethical considerations, transparency, and continuous learning.	No Public Evidence	Achieved	Achieved	Achieved	In Progress
ISO/IEC 27701	A standard that specifies requirements for establishing, implementing, and maintaining a Privacy Information Management System (PIMS). It serves as an extension to ISO 27001 for managing personal data, helping organizations comply with privacy regulations like GDPR.	No Public Evidence	No Public Evidence	Achieved	Achieved	No Public Evidence
HIPAA/BAA	The HIPAA Privacy Rule requires a signed contract, or Business Associate Agreement (BAA), to ensure a vendor will appropriately safeguard protected health information.	BAA available under some circumstances	HIPAA-ready Enterprise Offerings available since Jun. 26	BAA available for certain offerings	HIPAA-ready offerings available	HIPAA Business Associate Agreements (BAAs) for Enterprise customers
FedRAMP	A U.S. government-wide program that provides a standardized approach to security assessment, authorization, and continuous monitoring for cloud products and services.	Agency authorization in progress for a Class D (High) certification	Class D (High) certification	Class C (Moderate) certification	Gemini Enterprise and Gemini App: Class D (High) certification Gemini for Government: Class B (low) certification	No Public Evidence
GDPR	A legal framework that sets guidelines for the collection and processing of personal information from individuals in the EU. It grants individuals rights over their data and imposes strict rules on data transfers.	Aligns with GDPR in areas including data processing, security, and transfers	Aligns with GDPR in areas including data processing, security, and transfers	Aligns with GDPR in areas including data processing, security, and transfers	Aligns with GDPR in areas including data processing, security, and transfers	Aligns with GDPR in areas including data processing, security, and transfers
CSA STAR	A publicly accessible registry of the security and privacy measures that cloud computing firms offer.	STAR Level 1 (Self-Assessment)	STAR Level 2 (Third-Party Audit)	STAR Level 1 (Self-Assessment)	STAR Level 1 (Self-Assessment) and STAR Level 2 (Third-Party Audit)	No Public Evidence

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Source: Citi Research, Company Reports, FedRamp.gov, Cloud Security Alliance

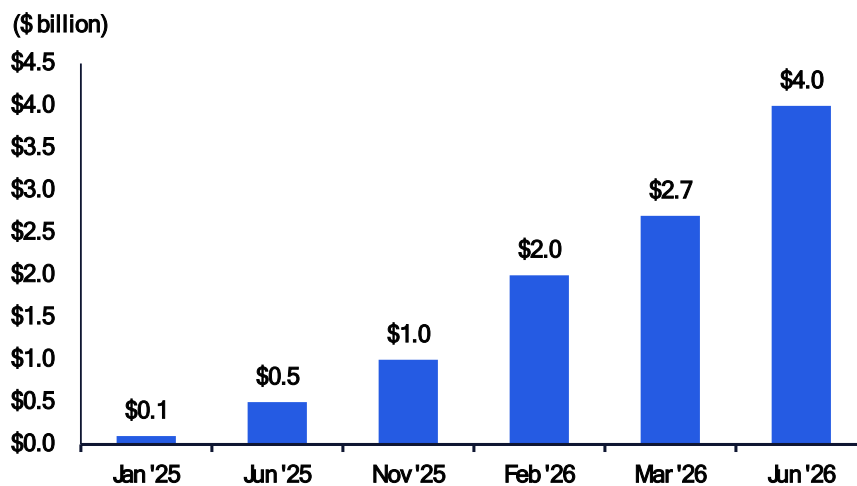
4) Cursor acquisition bolsters enterprise AI outlook and adds strategic value

In addition to providing an immediate ~\$4B ARR lift to SPCX's top line, the **Cursor acquisition (see announcement)** should improve SpaceX's standing in the AI landscape given Cursor's position as one of the frontier AI coding platforms, its high-caliber AI talent, and an established GTM playbook with significant room for growth in the enterprise. Alongside the option agreement to acquire Cursor, the two parties signed a compute agreement in April giving Cursor access to train its Composer coding model on Colossus clusters in exchange for SPCX gaining access to Cursor personnel and IP. We highlight the following three sources of strategic value:

Strategic value #1: Leading AI coding & agentic platform

The Cursor acquisition is a key opportunity for SPCX to close the gap with the predominant US frontier labs, as the two will partner to advance Grok and potentially develop jointly owned coding and knowledge work AI models, trained on SpaceX's compute infrastructure. Cursor's rapid ARR growth serves as a proof point for the overwhelming demand for AI-based productivity tools, starting with coding. The company crossed \$4B+ ARR as of early June 2026, growing nearly 50% compared to \$2.7B in March. Cursor's ARR performance demonstrates a proven, sticky developer revenue base that is competitive with other top tier coding tools such as GitHub Copilot. SPCX management believes that access to Cursor's data for future Grok model training runs will enable them to remain competitive as a frontier model and accelerate their competitive position in enterprise solutions. Cursor has over 10 trillion tokens of private coding data, more than GitHub, according to SPCX management. The company believes that developing a strong coding AI that can be scaled to more generalized tasks is the right approach to capture market share. On May 25, SpaceXAI released Grok Build (in beta), a coding agent and CLI for professional software engineering and complex coding work, a release signaling its more aggressive move into enterprise agentic workflows. Notably, Grok Build is only available to SuperGrok Heavy subscribers, which costs \$300/month (as of Jun. 29, there is a 67% discount offer for 3 months at \$99/mo. which then goes to \$300/mo.), compared to the standard SuperGrok subscription of \$30/month. The pricing could be prohibitive to broader near-term adoption, given similar AI coding agent functionality is bundled and offered on cheaper \$20/month plans for other frontier AI lab offerings.

Figure 33. Cursor ARR History



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Source: Citi Research, The Information

Strategic value #2: AI talent

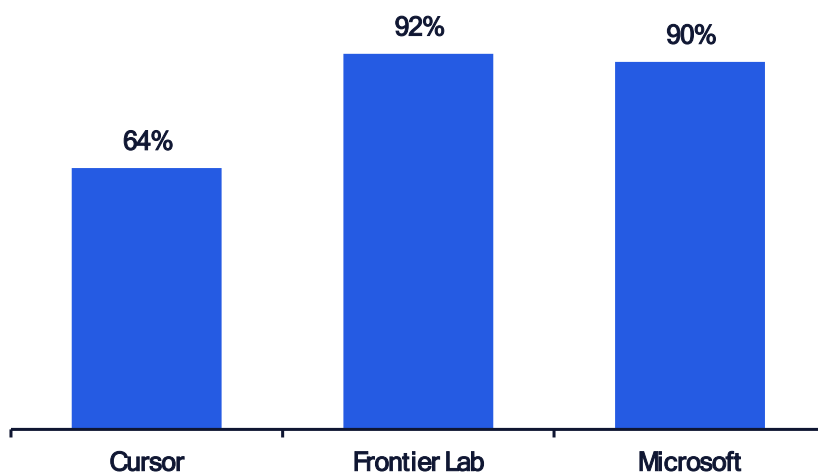
The Cursor acquisition fills an important talent and senior leadership gap caused by the departures of many technical staff and xAI's 11 other co-founders as of March 2026 (seven departures in 2026 alone) as a result of the SpaceX merger announcement with xAI and subsequent restructurings/reorganizations that took place (TechCrunch, Mar. 28, 2026). Cursor's headcount has expanded from 15

people two years ago to now over 700. In addition to the sheer engineering talent added, Cursor is expected to accelerate SPCX's goal of building the best coding models and agentic products. Cursor leadership first joined xAI in mid-March with the hiring of Andrew Milich and Jason Ginsburg, two senior product engineering leaders from Cursor. The release cadence of Grok products and feature velocity has picked up substantially since the key hires and strategic collaboration announcement between the two companies in April. SpaceXAI released Grok Speech-to-Text and Text-to-Speech APIs, Grok Voice Think Fast 1.0 voice agent in April, Grok Imagine Quality Mode API, connectors/integrations/skills, Grok Build CLI based coding agent and Grok Build 0.1 coding model in May, and Grok Imagine 1.5 Preview, Grok Imagine Video 1.5, a plugin marketplace, agent dashboard, and Grok for PowerPoint in June.

Strategic value #3: Cursor a GTM entry point into enterprise

Cursor provides SpaceXAI with a better entry point into the enterprise. Despite Cursor's largely bottoms-up and developer-driven sales motion, it has been able to scale to over 50K customers within three years of launching its AI-powered code editor, reaching 64% penetration of the Fortune 500 and a couple thousand enterprise customers. This is still lower than large tech incumbents like Microsoft, with over 230K customers and 90% Fortune 500 adoption, and Google, with 75% of GCP customers utilizing Google's AI products, suggesting plenty of enterprise growth potential ahead.

Figure 34. Cursor's Fortune 500 Penetration vs. Other AI Platforms



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Source: Citi Research, Company Reports, Thomson Reuters

With Project Origin, Cursor is expanding its offerings to include more enterprise-oriented features such as code repositories for storing software projects, security reviews, and software integrations, features commonly found on GitHub. While Cursor already overlaps with many of the capabilities of GitHub, we think at least half of the latter platform's low-single-digit billion dollar revenue run-rate represents a replacement opportunity that Cursor can go after (GitHub's annual revenue run-rate was estimated to be \$2B at the end of 2024 per Business Insider, Jan. 2025). We believe Cursor and SPCX joint progress on agentic tooling and AI

coding models (Grok Build, Composer 2.5) and focus on providing the lowest cost software tools could create a compelling GTM narrative for the enterprise. In terms of coding agents, Cursor's Composer 2.5 Fast model averages a cost per task of \$0.55, roughly 90% below the cost per task of other leading US frontier coding agents, whereas the performance differential is about 30%, per Artificial Analysis' Coding Agent Index rankings (Artificial Analysis, Jun. 28, 2026).

Figure 35. Cursor's Customer Use Cases

Company	Industry	Location	Bugbot Used (Y/N)	KPI 1	KPI 2	KPI 3
Amplitude	Software	North America	Y	60 - 70% of low-risk PRs merged directly to production	1000+ automated agent runs every week without manual prompting	3x increase in weekly production commits since adopting cloud agents
PayPal	Financial Services	North America, EMEA, Asia Pacific	N	40% potential increase in capabilities delivered	6x faster Java upgrade across 3,000 applications	Daily deployment cadence, up from monthly release cycle
Coinbase	Financial Services	North America	N	90% reduction in time from idea to production	55% increase in PRs merged per engineer. 75% of PRs created by agents, across cloud and local	7 hours of work saved per week per engineer
Box	Software	North America	N	30% - 50% roadmap throughput increase	85% Box engineering adoption	80% faster React migration
DropBox	Software	North America	N	550,000 files indexed in Dropbox's monorepo	1M+ Cursor-generated lines accepted monthly	90% engineering adoption
Faire	E-commerce	North America	N	2x increase in weekly PR throughput, driven by cloud agents	18 month migration for a full team reduced to one engineer and a fleet of agents	2000+ automated agent runs per week without any manual prompting. 25+ Cursor Automations for bug fixes, CI investigations, and code review
Stripe	Financial Services	North America	N	3,000+ Stripe developers using Cursor	5-year high in developer tooling sentiment	
Wayfair	Retail	North America	N	94% reduction in tag-validation model inference cost during the December hackathon. 90% further cost reduction in the March hackathon against the new December baseline	110 distinct model variants tested by 5 researchers in 2 days	20+ parallel Cursor agents run per researcher during experimentation sprints
Money Forward	Financial Services	Asia-Pacific	N	1,000+ Money Forward employees using Cursor	15-20 hours saved weekly per engineer	70% faster QA test generation
National Australia Bank	Financial Services	Asia-Pacific	N	6,000+ Cursor users at NAB, scaling toward 10,000+	3x faster legacy migration and refactors for banking applications	3 weeks to build a greenfield payment app instead of a four-month scope
NVIDIA	Semiconductors	North America	N	30,000 NVIDIA employees using Cursor	3x more code committed when using Cursor	Faster junior developer onboarding
PlanetScale	Cloud Infrastructure	North America	Y	80% Bugbot resolution rate	2,000+ PRs reviewed monthly	2 FTEs dev time saved
Salesforce	Software	North America	N	75% Salesforce engineering adoption	30%+ PR velocity increase	85% less time on legacy test coverage

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Source: Citi Research, Company Reports

5) Enterprise AI risks

While SpaceX's acquisition of xAI in early 2026 integrates frontier AI capabilities into its vertical stack, the Grok Enterprise model faces substantial risks of underperforming compared to established market leaders in the enterprise AI market. The development of highly complex frontier models like Grok requires access to a limited pool of elite engineering and technical talent, a resource that is subject to intense, multi-front competition. Because key employees are not bound by long-term employment agreements, the AI segment remains highly vulnerable to sudden leadership and technical attrition, which could compromise its rapid iteration speed. This talent risk is further exacerbated by the extreme selectivity required to build high-performance systems, alongside the high training costs of onboarding personnel before their retention value is realized. Beyond talent constraints, xAI is navigating severe legal and operational headwinds, including multiple class-action lawsuits regarding Grok's image generation features and allegations of regulatory violations. If these talent acquisition, retention, and litigation risks materialize, they could degrade Grok's competitive position, leading to operational bottlenecks and substantial financial liabilities that hinder its enterprise monetization.

Unlocking Trillion-Dollar Market Opportunities: Orbital AI

- 1) Credible, Yet Execution-Dependent Extension of Launch, AI, & Connectivity
- 2) Addressing the Constraints in the Terrestrial AI Market
- 3) Economics Attractive on the Non-Chip Infrastructure
- 4) Orbital AI Risks

1) Credible yet execution-dependent extension of launch, AI, and connectivity

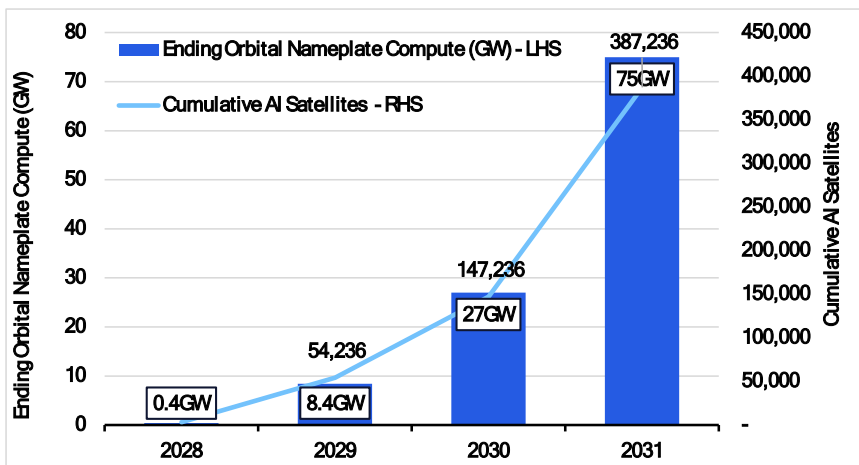
The company's orbital AI compute vision (SpaceX, Feb. 2026) is a credible yet execution-dependent extension of the company's core advantages in launch, satellite manufacturing, Starlink networking, and vertically integrated hardware. SpaceX will control both the launch vehicles required to deploy AI satellites as well as the manufacturing capabilities needed to build them. Given SpaceX's long lead in launch capabilities and operational proficiency in satellites, SpaceX's owns more of the prerequisite stack for AI satellites than any other likely competitor.

Is space-based AI infrastructure inevitable? Critical to the company's view is a belief that current advances in AI across both training and inference will require an amount of electricity that cannot be met with terrestrial methods. Therefore, in the long term, space-based AI is the solution to the widening gap between compute demand and power availability. In other words, "try getting the permits [to cover Nevada in solar panels]...it's harder to scale on ground than it is to scale in space" (A Cheeky Pint, Feb. 2026). With this perspective in mind, a key debate becomes not whether the company can successfully put compute in orbit – we view the technological hurdles as either solved or able to be solved in due time – but whether orbital compute can scale to the level of cost, reliability, power density, and utilization that renders AI satellites competitive with terrestrial alternatives before there is material progress in resolving terrestrial infrastructure bottlenecks by hyperscalers, utilities, nuclear / gas generation, or other off-grid architectures. Accordingly, many view timing as the critical variable in the orbital AI investment thesis, the primary determinant of which will be progress against Starship achieving rapid reusability. That said, if demand for compute is truly as insatiable as recent data points suggest, SPCX argument of scalability at the extremes holds merit unto itself.

We see four linked suppositions that underpin SpaceX's differentiated argument: (1) launch costs should decline structurally with Starship rapid reusability; (2) orbital solar power removes the largest long-term operating cost for terrestrial solutions (i.e. power); (3) radiative cooling reduces thermal opex; and (4) Starlink's optical mesh provides the networking substrate needed to turn satellites into effective distributed AI clusters. Of these four, we anticipate that launch costs and optical mesh will be the engineering challenges that command the most focus, particularly since the implied launch cadence required to meet management's goal of 100 GW per year necessitates till now unwarranted regulatory approvals, and the downlink speeds required to mitigate latency expected for inference could prove

insufficient. We expect the company to reach 400MW of installed nameplate compute by 2028, growing to 75GW in 2031 under these conditions.

Figure 36. Orbital AI Compute Build-Out: Installed Capacity and Satellite Count



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Source: Citi Research

2) Addressing constraints in the terrestrial AI market

The constructive case for orbital AI is that it addresses the most constrained parts of the terrestrial data center stack: **power, cooling, permitting, grid interconnection, and physical build time**. These satellites will be enabled by solar power (free and clean), Starship deployment, Starlink optical networking, and high-volume satellite manufacturing. Early AI satellite power targets of 100 kW per satellite, compute-density targets of 80-120 kW per ton, and an aspirational deployment rate of 100 GW per year would require ~1M metric tons per year to orbit and the company aims to deploy up to a million satellites (FCC, Jan. 2026). Accordingly, Starship is central to the company's ambitions as it will be the only fully reusable system designed to carry 100+ metric tons to orbit, which is the style of launch architecture needed for 200 kW-class satellites and gigawatt-scale deployment.

However, although there are proof points, none of this has been done at a commercial scale. Starcloud, an orbital data center startup, has shown that orbital AI compute is technically possible at demo scale, launching Starcloud-1 with an NVIDIA H100 and claiming the first LLM training/fine-tuning in orbit (LinkedIn, Nov. 2025). Launched by a Falcon 9 and the size of a small refrigerator, the H100 ran inference on a Google Gemma model and trained Andrej Karpathy's nano-GPT on the complete works of Shakespeare (X, Dec. 2025). This technical milestone proved that a data-center-class GPU could survive launch, operate on solar power in orbit, and execute LLM inference and small-scale training workloads. However, scalable power density, thermal rejection, high-bandwidth networking, maintenance/replenishment, and economics remain unproven.

In our view, orbital AI is less about whether it is possible (it is) and more about whether the technology hits the right ratio of performance / cost. The technology building blocks increasingly exist, the question remains whether SPCX can integrate them at data center scale and at a cost that outperforms terrestrial alternatives. Continuing to diligence orbital AI satellite performance as technology and cost curves evolve will be a focal point for some time.

3) Economics attractive on the non-chip infrastructure

We view orbital AI as potentially disruptive to the non-chip portion of the AI data center stack first, and to the full cost-per-token stack only later if Terafab materially reduces accelerator costs. In the interim, AI satellite economics are primarily an ex-chip infrastructure advantage, though our analysis demonstrates that the cost curve remains an engineering challenge. The economics can become attractive on the non-chip infrastructure layer if Starship cadence and satellite manufacturing scale materialize. However, the processor layer still remains a critical cost bucket, so orbital AI does not become a full-stack cost advantage unless SpaceX also reduces accelerator cost through Terafab, proprietary chip design, or materially better utilization. The company is targeting 90% value capture through its supply chain derisking effort via Terafab.

We estimate that 30% of terrestrial AI data center costs are dedicated to construction/shell/MEP/grid interconnection, with the remaining 70% dedicated to chips and other critical IT equipment. Therefore, SpaceX's current cost advantage could remain structurally bounded if it only reduces the construction, power, cooling, and deployment layers while still utilizing higher cost, NVIDIA-class silicon. Our work suggests that relatively aggressive assumptions around launch cost reduction, satellite manufacturing learning curves, solar array costs, and thermal simplification are needed to justify the longer-term costs. Our illustrative analysis assumes a 200 kW AI satellite, 2-ton launch mass, 100 kW/ton specific power, 190 GPU-equivalent chips, and 10% chip infant mortality & non-compute power distribution, yielding 180 kW effective delivered power after adjustment. On that basis, the model estimates ex-chip, ex-Terafab total satellite cost of \$8.5M in 2029E, falling to \$6.5M in 2030E and \$5.8M in 2031E, equivalent to \$47/W, \$36/W, and \$32/W upfront, or \$9.4/W/year, \$7.2/W/year, and \$6.4/W/year over a five-year useful life.

4) Orbital AI Risks

Given this is a technology that has never been attempted at scale before, there is a possibility that the architecture is economically fragile once real-world losses materialize. These losses could include liquid cooling loop mass, additional shielding, dedicated radiator mass, radiation-hardening premiums, solar cell cost inflation versus terrestrial benchmarks, and failure related derating. Thermal management may become an important technical gating item. SpaceX's AI1 concept describes a deployable liquid radiator with redundant pumping loops, suggesting that the company is already contemplating active thermal systems rather than purely passive radiative cooling – though this could be describing a chip-level heat transport loop which works simultaneously with a radiative rejection mechanism. Finally, we note that there are several orbital governance considerations, including debris, collision-avoidance, deorbit, spectrum, astronomy, and optical brightness.

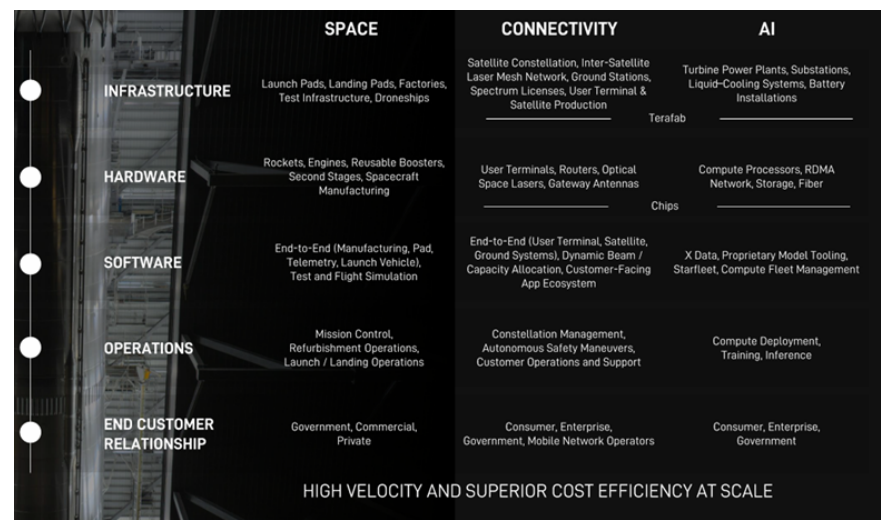
Extreme Vertical Integration

- 1) High Velocity and Superior Cost Efficiency at Scale
- 2) Overview of Terafab and Extreme End-Customer Vertical Integration
 - A) Why Terafab?
 - B) How Big Does Terafab Need to Be?
- 3) Terafab Implications to the Semiconductor Industry
- 4) SpaceX Partnership with Tesla

1) Extreme vertical integration explained

SPCX believes one of its strengths is “Extreme Vertical Integration” which enables the company to operate at uniquely high velocity and super cost efficiency at scale. SPCX sees itself as pursuing a “shovels to tokens” strategy in scaling AI compute rapidly and efficiently. This first-principles, manufacturing philosophy rejects conventional industry assumptions in favor of rapid in-house prototyping and development. More specifically, by integrating design, engineering, manufacturing, and operations all under a single organizational umbrella, the company effectively bypasses the fragmented supply-chain challenges faced by legacy competitors in multiple business lines. This nimble framework is systematically reinforced across all business segments through the rigorous application of optimization practices to eliminate operational bottlenecks and speed design cycles.

Figure 37. SpaceX Vertical Integration

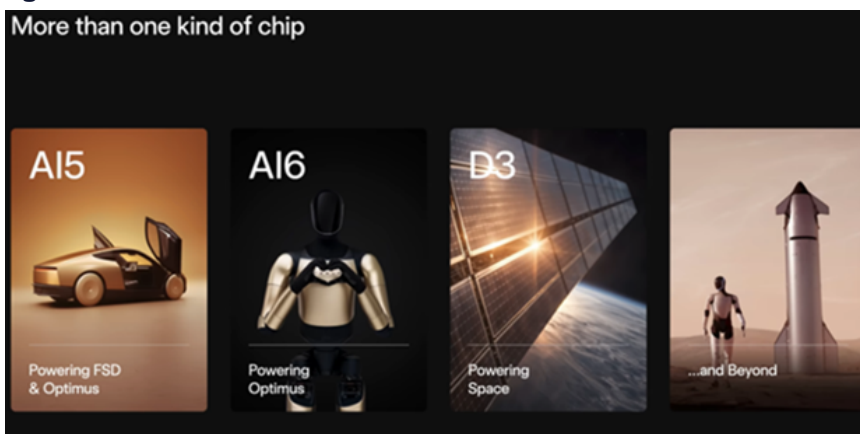


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Source: Citi Research, Company Reports

2) Overview of Terafab

Terafab is a multi-year, capital-intensive chip manufacturing initiative led by SpaceX in partnership with Tesla and Intel, announced in March 2026 (Tesla collaboration) and expanded in April 2026 (Intel joining). The vision is to build what could become the world's largest AI chip manufacturing ecosystem, targeting one terawatt of annual compute output. Terafab is architected as a fully vertically integrated, "closed loop" facility that combines lithography mask design, fabrication of logic and eventually memory chips, advanced packaging, and testing within the same plant. This closed-loop approach allows chips to be designed, manufactured, tested, and quickly refined without relying on external feedback cycles, dramatically accelerating design improvements and yield optimization. SpaceX leads chip architecture for orbital AI and deployment and brings prior in-house silicon design experience, Tesla contributes chip architecture for edge inference (Optimus, FSD) and provides the volume demand needed to justify fab economics, and Intel provides core manufacturing technology support for both foundry and packaging. The strategy is to simplify manufacturing by focusing on ~2–3 core chip types, while still complementing – not replacing – continued reliance on third-party suppliers.

Figure 38. Terafab for Tesla and Orbital AI



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Source: Citi Research, Company Reports

Tesla partnership. While SpaceX and Tesla have not finalized the details related to the collaboration, we believe that Tesla can contribute across several dimensions: 1) serve as a cornerstone customer, contributing to its in-house chip architecture and providing the high volume, vertically integrated demand needed to underpin fab utilization and economic viability; 2) provide financial support and capex investment, as signaled in its early 2026 earnings commentary that Tesla's current \$20Bn+ capex guidance for 2026 "does not include potential investments in solar cell manufacturing or our Terafab as we're still in early phase. And we plan to provide an update in future quarters"; and 3) offer technical and operational support. Tesla currently has more than 20 job postings related to Terafab, spanning program management, facilities and process engineering that covers more than 10 semiconductor manufacturing process steps, and automation/software engineering, collectively pointing to an active build out of capabilities required to

design, construct, and operate a vertically integrated semiconductor fabrication facility.

As background, Tesla's AI5 is a next-gen edge chip for its cars and robots, focused on efficient real-time inference for FSD and Optimus with better performance per watt and memory bandwidth vs current version. It is expected to enter sample production in 2026 with mass production in 2027. AI6, as an evolution of AI5, is a more advanced, unified compute platform that can support both inference and some training workloads and will be used for edge/inference cases like Optimus as well as for data centers. It is expected to enter mass production in 2028. Current generation of Tesla's chip (AI4) is manufactured at Samsung, with the expectation that AI4/5 will be dual sourced from both Samsung and TSMC. Meanwhile, Dojo was initially deprioritized as Tesla shifted toward external GPUs and focused on in-vehicle inference. It has since been revived as "Dojo 3", or D3/AI7, and will be heavily dedicated for space-based AI compute.

Intel partnership. We believe Intel's role in the partnership is to provide leading-edge foundry expertise, including its FinFET-based roadmap and upcoming 18A/14A GAA process nodes, alongside deep know-how in advanced semiconductor production. It also brings advanced packaging capabilities such as EMIB and Foveros, which have been gaining traction in the industry, as advanced packaging is increasingly critical for modern AI chips given performance gains depend more on tightly coupling multiple chiplets, improving interconnect density, and overcoming reticle size limits rather than relying solely on node shrinks. In addition, Intel supports the ecosystem through PDKs, manufacturing integration, and close co-optimization with chip designers to enhance performance, power efficiency, and yield. As a US based leader in advanced node manufacturing, Intel further strengthens the partnership with a resilient domestic supply chain and the ability to support complex applications across AI data centers and edge AI applications. On the other side, while Intel has historically participated in memory manufacturing across DRAM, NAND, and its own 3D XPoint (Optane) technology, it no longer operates as a major memory producer today and instead focuses on embedding, integrating, and optimizing memory within advanced AI systems.

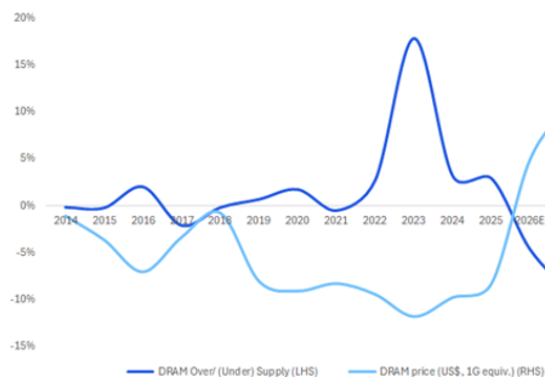
A) Why Terafab?

Terafab functions as both a supply chain de-risking and margin capture strategy, addressing what SPCX sees as one of the key constraints to AI scaling: chip supply, data center infrastructure, and power. By insourcing chip manufacturing, SpaceX can reduce reliance on constrained external suppliers, secure access to significantly larger volumes of compute hardware, and capture value currently spread across foundries, merchant GPU vendors, and memory providers. It also enables custom chip optimization for both space and terrestrial applications, while the closed-loop development model accelerates iteration and performance gains. Although SpaceX will continue to procure chips from third parties, Terafab provides a long-term pathway to scale compute far beyond current supply limits and strengthen control over the entire physical AI stack.

It is well understood by now that the semiconductor industry is facing persistent shortages across logic and memory driven by insatiable AI demand. From the logic perspective, TSMC faces prolonged tightness in advanced nodes (specifically N3 and N2) and advanced packaging (such as CoWoS), as large die-size AI GPUs and accelerators migrate to these cutting-edge processes faster than capacity can be built. Although TSMC is aggressively expanding facilities in Taiwan and Arizona

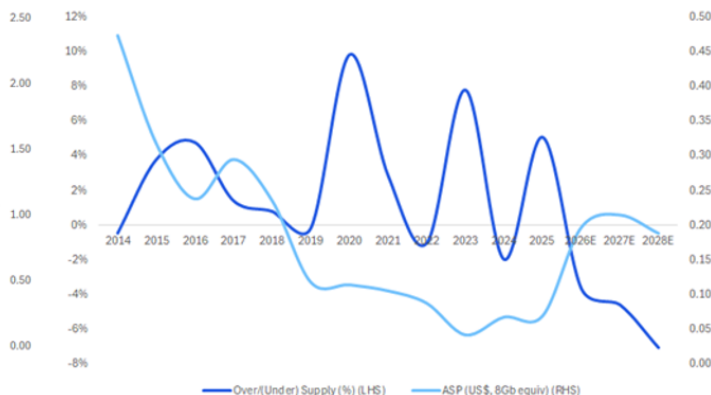
to narrow the gap, the long 2–3 years lead times for fab construction mean supply for leading-edge nodes will remain structurally tight for the foreseeable future. Simultaneously, the memory market is experiencing an unprecedented surge in demand for HBM and advanced DRAM. Memory leaders such as Micron and Samsung have indicated they cannot fully meet broad-based customer demand, leading to strict supply allocations and rising prices across both high-end and conventional memory categories.

Figure 39. DRAM Supply/Demand & ASP



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Source: Citi Research

Figure 40. NAND Supply/Demand & ASP



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Designing its own custom silicon and insourcing manufacturing have the potential to drive substantial cost savings by eliminating multiple layers of high-margin intermediaries in today's semiconductor value chain, such as ~60% gross margin at advanced foundries like TSMC, 75% gross margin at GPU vendors like Nvidia, and 80% gross margin at memory players like Micron (or 30% normalized level). Assuming 1GW AI factory costs around \$50Bn, including 50% GPU/ASICs, 20% memory, and rest from storage, networking and other infrastructure related costs, internalizing chips design/manufacturing/memory could save up to ~\$19Bn/\$3.5Bn/\$8Bn for SPCX per GW, or nearly 60% of cost.

B) How big does Terafab need to be?

Terafab is conceived at a scale that would far exceed anything in today's semiconductor or AI infrastructure landscape. With a target of ~1 terawatt of annual compute output spanning ~100 million square feet, it would be roughly 50× larger than current global AI chip production of ~20 GW, per Elon Musk, effectively rivaling or surpassing the entire existing industry in itself. A court filing (see [filing](#)) shows Terafab would require about \$55Bn in initial investment for its first phase in Texas, with total spending potentially rising to around \$119Bn for the full buildout.

3) Terafab implications to the semiconductor industry

Citi estimates that total peak power demand for Space X (including both Orbital and terrestrial) could go up to ~90GW compute by 2031. To achieve this output level, a manufacturing cluster needs to have a total output of ~260k wspm (assuming 500w TDP and 500mm² die size), which could translate into an accumulative capital investment of at least ~\$320Bn+ through 2031. Putting this

required scale into an industrial context, we benchmark against major leading-edge fab buildouts in the US by TSMC, Samsung, Intel, as well as newer entrant Rapidus who is a new foundry player and is building an advanced node fab in Japan. TSMC first announced a \$12 billion investment in Arizona in 2020 for a single fab producing 5nm-class technology with 20k wspm capacity, then expanded to \$40 billion by adding a second fab, and later to \$65 billion with the addition of a third fab for N4, N3 and eventually N2/A16 technologies. In March 2025, TSMC announced a further \$100 billion expansion, bringing total committed investment to \$165 billion. The plan now encompasses a roughly 1,100-acre (~47.9 million sq ft) campus, consisting of six wafer fabs, two advanced packaging facilities, and one R&D center, creating an integrated US “Gigafab” with industry estimated capacity of approximately 100k wafers per. Longer term, media reports suggest TSMC is evaluating a larger buildout involving 12 fabs and four packaging plants, which could lift total investment above \$265 billion and capacity toward over 200k wspm. Samsung first announced its Texas Taylor fab in 2021 with \$17 billion investment on a ~1,200-acre (~52.3 million sq ft) campus and expanded in 2024 to ~\$37 billion total, covering 2 fabs and 1 R&D facility. Intel’s Ohio “Silicon Heartland” fab was announced in 2022 with a ~\$20 billion plan for two leading-edge fabs on a ~1,000-acre (~43.6 million sq ft) site, later expanded (2022–2023) to >\$28 billion, with a long-term goal of up to eight fabs plus supporting infrastructure. It is common to take at least 3–4 years before a sizable greenfield fab from groundbreaking into production in the US, though Terafab could be accelerated with the help of Intel’s prior experience from the Ohio fab.

4) SpaceX partnership with Tesla

SPCX specifically discussed its unique relationship with Tesla in its S1. As background, the SpaceX–Tesla collaboration evolved meaningfully following Tesla’s January 2026 commitment to invest in xAI, which was converted into an equity interest in SpaceX upon SpaceX’s acquisition of xAI. This equity alignment creates a foundation for Tesla and SpaceX to evaluate future strategic opportunities between the companies, building upon their longstanding collaborative relationship. The partnership addresses what SpaceX identifies as the key constraints in AI growth: chip manufacturing, data center infrastructure, and power generation, with SpaceX believing it is better positioned than other AI companies given its unique control over the full physical stack. SpaceX’s strategic view is that the future of AI will be determined by control of the physical stack, as the key constraints in continued AI growth are physical rather than algorithmic. The collaboration enables both companies to explore areas of strategic cooperation that leverage their complementary capabilities in manufacturing, computing, and infrastructure. This partnership model demonstrates SpaceX’s approach to building aligned relationships that can accelerate innovation and create competitive advantages across multiple technology domains. We expect the unique relationship between SPCX and Tesla to grow closer over time.

Bull/Bear: Space Exploration Technologies (SPCX.O)

US\$ **366.00**



▲128% Upside

US\$ **200.00**



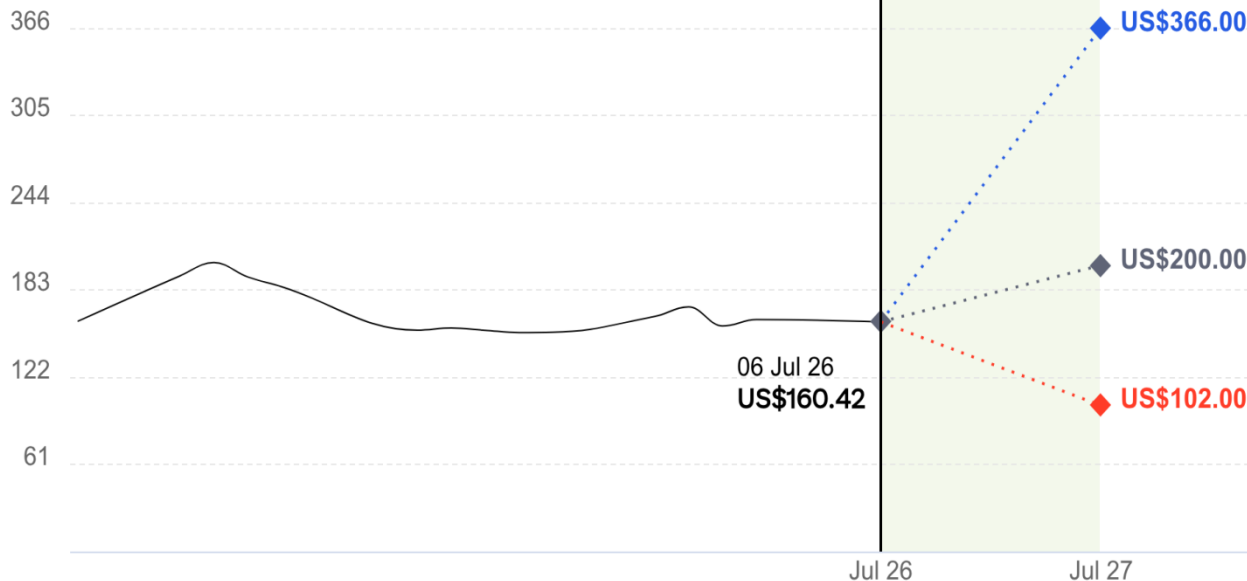
▲25% Upside

US\$ **102.00**



▼36% Downside

US\$



Spread 165pp

Current Price and expected returns (upside/downside) as of 06 Jul 2026

BULL Assumptions



- Out bull case is predicated on the demonstration of rapid reusability of Starship ahead of schedule and the successful implementation of AI orbital satellites, driving outer year compute forecasts.

BASE Assumptions



- We derive our \$200 Base Case Price Target using the avg of three methods, 1) EV/Rev:Growth & EV/EBITDA:Growth vs. Trillion \$ Comps on 2027 est, 2) SOTP on 2027 estimates, and 3) EV/Rev & EV/EBITDA vs. Trillion \$ Comps on 2030 estimates.

BEAR Assumptions



- Our bear case assumes a delay in the company's launch forecast and the implementation of orbital AI satellites, pushing our forecast to the right.

Space Exploration Technologies

Company description

SpaceX is a vertically integrated technology leader building the foundational hardware and software infrastructure for the future of space, global connectivity, and artificial intelligence. The company's Launch segment pioneered full reusability to provide affordable access to orbit, while its Connectivity segment operates the world's largest satellite constellation to deliver high-speed internet globally. Following its acquisition of xAI, SpaceX has also established a dedicated AI segment that integrates frontier models like Grok with massive terrestrial and orbital compute infrastructure.

Investment strategy

We rate shares of SPCX a Buy due to the following points: 1) Unrivaled launch capabilities right now and for the foreseeable future; 2) Unique capability of scaling space infrastructure and unlocking trillion-dollar market opportunities (Orbital AI and Starlink); 3) Extreme vertical integration, which reinforces the company's ability to drive costs down, throughput up, and deliver at a scale at which no competitor can replicate; and 4) Differentiated margin and growth profiles.

Valuation

We derive our \$200 base-case price target for SPCX using the average of three valuation methods: 1) 2027E growth adj. multiples (EV/Revenue/Growth and EV/EBITDA/Growth) for the "Trillion-Dollar Comps" (a basket of similar trillion-dollar+ market caps); 2) sum-of-the-parts (SOTP) valuing Space, Connectivity, and AI separately using 2027E growth adj. multiples (EV/Revenue/Growth and EV/EBITDA/Growth) for industry-specific comparable companies; and 3) 2030E comparable company multiples (EV/Revenue and EV/EBITDA) for the "Trillion-Dollar Comps" (a basket of similar trillion dollar+ market caps).

Risks

Risks to our thesis, which could impede the shares from reaching our price target, include headwinds to the launch segment which reference the failure to demonstrate rapid reusability on Starship, lack of launch infrastructure, and regulatory headwinds from the FAA and other government organizations. The failure to rapidly gain market share in Starlink mobile, as well as demonstrate the ability to create AI orbital satellites capable of compute, also presents risks to SPCX.

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Appendix A-1

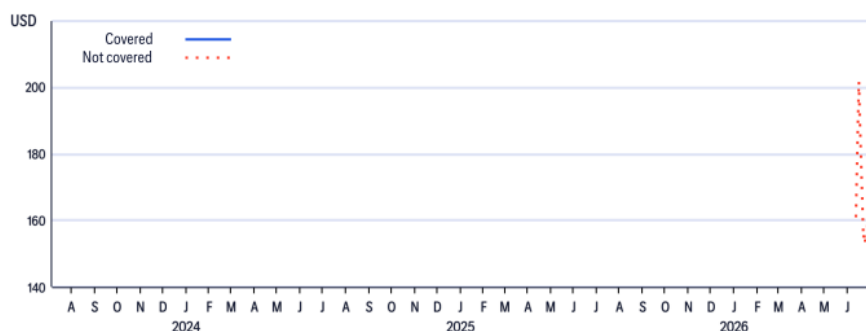
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Fundamental Research



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